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Research Discussion: How to do Research

(Research Questions, Plan, Implementation, and Performance Evaluation)

Vinh Dinh Nguyen
PhD in Computer Science

Outline

- **How to read a research paper**
- **How to develop research questions**
- **Study and Implement Existing Methods**
- **How to develop a research plan**
- **Implement Your Proposed Method**
- **Performance Evaluation**
- **Enough Contributions for Submission**

Mentor's Information



Vinh Dinh Nguyen ✎

FPT University Can Tho Viet nam

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Artificial Intelligence Machine Learning Computer vision Pattern Recognition

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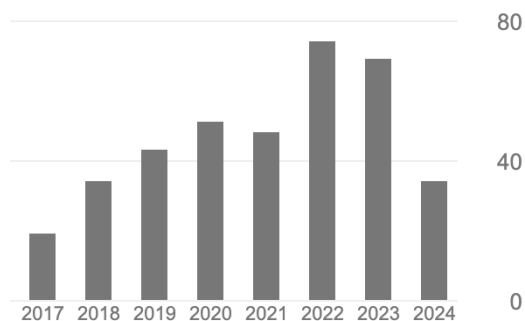
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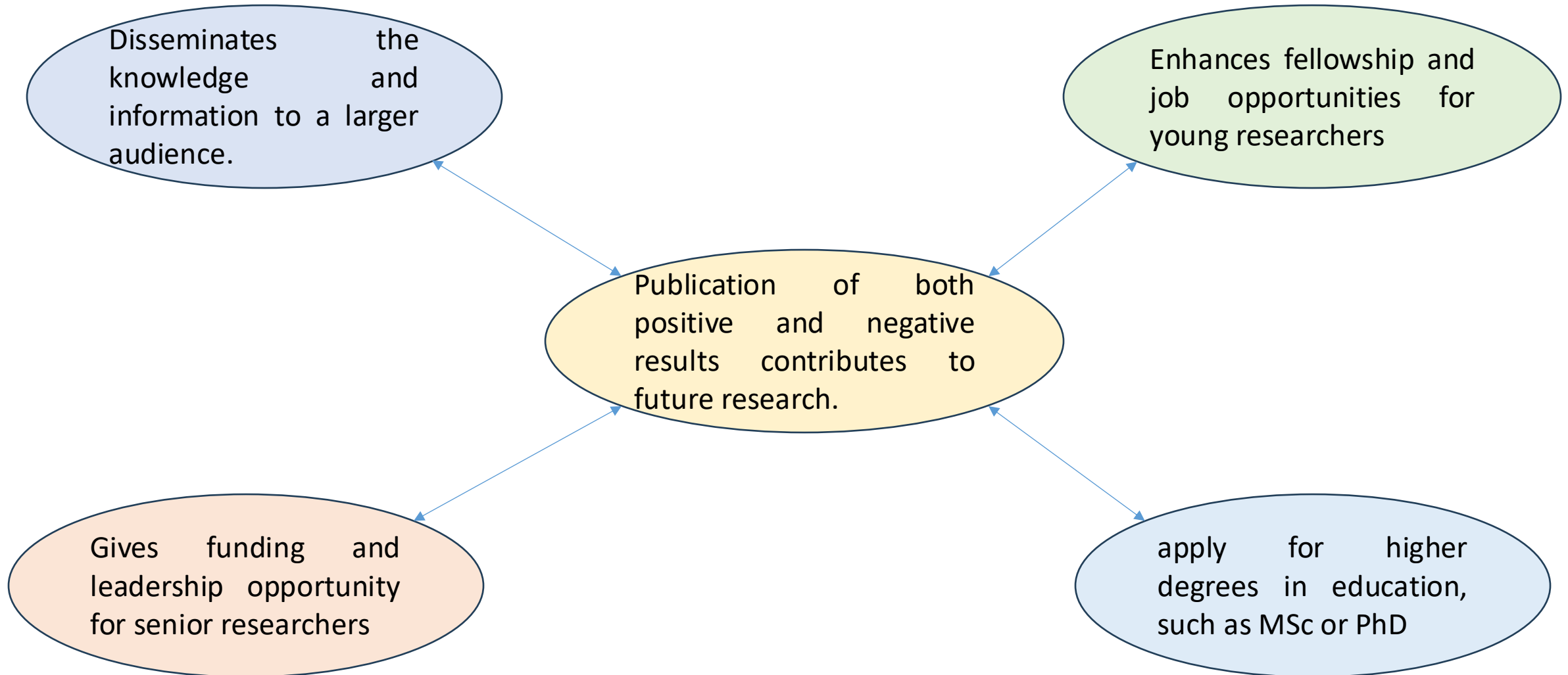


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Write a scientific research paper: Why?



Outline



➤ **How to read a research paper**

➤ **How to develop research questions**

➤ **Study and Implement Existing Methods**

➤ **How to develop a research plan**

➤ **Implement Your Proposed Method**

➤ **Performance Evaluation**

➤ **Enough Contributions for Submission**

How to Read a Research Paper

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, VOL. 24, NO. 2, FEBRUARY 2014

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Step 1:

Read Paper Title and answer the following questions?

1. What is the research topic?
2. Is it closed to your research interest?
3. Is it a new research approach?
4. General idea of what the research is about
5. Does the title use keywords that are relevant to the topic?
6. Does the title catch your attention or pique your interest in the research?

Support Local Pattern and Its Application to Disparity Improvement and Texture Classification

Vinh Dinh Nguyen, Dung Duc Nguyen, Thuy Tuong Nguyen, Vinh Quang Dinh, and
Jae Wook Jeon, *Member, IEEE*

Abstract—The local binary pattern (LBP) and its variants have been widely investigated in many image processing and computer vision applications due to their robust ability to capture local image structures and their computational simplicity. The existing LBPs extract local structure information by establishing a relationship between the central pixel and its adjacent pixels. However, most LBPs miss the relationship among all of the pixels in the local region. Therefore, this paper proposes a novel model to establish this relationship by introducing a support LBP. The proposed model improves the performance of the existing LBP methods and results in lower sensitivity to illumination changes and radiometric variations. Moreover, the proposed model has been successfully investigated in two applications: disparity map generation and texture classification. For disparity map generation, the proposed model reduces the root mean square (RMS) error by 23.6% (in Baby1 dataset, Middlebury), and 16.58% (in Aloe dataset, Middlebury) as compared with the standard LBP under radiometric variation conditions. Moreover, the proposed model reduces the RMS by 28.11% as compared with the standard LBP under the Gaussian noise condition in the ESATS dataset. For texture classification applications, the proposed model improves the classification results from 96.26% to 98.13% on the Outex database, from 88.03% to 91.41% on the Xu database, and from 94.00% to 96.67% on the KTH-TIPS database as compared with the completed LBP.

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Texture classification is an application that is used to assign an unknown texture to the set of predefined textures. The assignment process can be based on the training of VZMR8, VZ_Joint [4], [5], or free training [6]. VZMR8 uses a predefined texon library that is constructed from a training set and a set of filter banks in order to classify the unknown texture. A texture dataset is divided into two sets, one of which is for a sample model and one that is used for testing during the free training. In conventional LBP [1], a uniform pattern is used to extract the texture information. It achieves very good accuracy in texture classification by combining multiple-scale LBPs, which work well in the case of straight edges or low curvature edges. However, this method is not successful in cases with

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Vinh Dinh Nguyen received the B.S. degree (*magna cum laude*) in computer science from Nong Lam University, Ho Chi Minh City, Vietnam, in 2007 and the M.S. degree in electrical and computer engineering from Sungkyunkwan University, Suwon, Korea, in 2012. He is currently working toward the Ph.D. degree with the School of Information and Communication Engineering, Sungkyunkwan University.

His research interest include computer vision, image processing, and graphics processing unit computing.

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How to Read a Research Paper

Step 3:

Read the abstract of the paper and answer the following questions?



1. What methods or approaches were used in the research?
2. What are the most important findings or results of the study?
3. Which techniques that you known/or unknown?
4. What conclusions or implications are drawn from the findings?
5. What is the overall purpose or goal of the research?

Abstract—The local binary pattern (LBP) and its variants have been widely investigated in many image processing and computer vision applications due to their robust ability to capture local image structures and their computational simplicity. The existing LBPs extract local structure information by establishing a relationship between the central pixel and its adjacent pixels. However, most LBPs miss the relationship among all of the pixels in the local region. Therefore, this paper proposes a novel model to establish this relationship by introducing a support LBP. The proposed model improves the performance of the existing LBP methods and results in lower sensitivity to illumination changes and radiometric variations. Moreover, the proposed model has been successfully investigated in two applications: disparity map generation and texture classification. For disparity map generation, the proposed model reduces the root mean square (RMS) error by 23.6% (in Baby1 dataset, Middlebury), and 16.58% (in Aloe dataset, Middlebury) as compared with the standard LBP under radiometric variation conditions. Moreover, the proposed model reduces the RMS by 28.11% as compared with the standard LBP under the Gaussian noise condition in the ESATS dataset. For texture classification applications, the proposed model improves the classification results from 96.26% to 98.13% on the Outext database, from 88.03% to 91.41% on the Xu database, and from 94.00% to 96.67% on the KTH-TIPS database as compared with the completed LBP.

How to Read a Research Paper

Step 4:

Read the conclusion of the paper and answer the following questions?

1. What are the main findings and results presented in the conclusion?
2. How do the findings answer the research question or address the research objectives?
3. Are there any unexpected or surprising results discussed in the conclusion?
4. Do the conclusions align with the research hypothesis or initial goals?
5. Are there any limitations or weaknesses of the study mentioned in the conclusion?
6. What are the practical implications of the findings?
7. Are there any recommendations or suggestions for future research provided?

V. CONCLUSION

This paper analyzed LBP from the viewpoint of the local structure in order to investigate the feasibility and effectiveness of the support binary pattern for various existing binary patterns and applications. The support binary pattern successfully establishes the relationship of a pixel from various directions in the local region. Robust experiments on two common applications, disparity map generation and texture classification, demonstrate that the proposed model improves the performance of current local pattern methods. The main contributions of this paper include the following.

- 1) Proposal of a new model to establish the relationships among all of the pixels in the local region (support local pattern).
- 2) Successful application of the proposed model to disparity map generation and texture classification to improve their performances in the case of radiometric variations and Gaussian noise.

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Conclusion

- 3) Easy application of the proposed model to other local patterns to improve performance, while preserving the simplicity of the original methods.

Support local pattern is a general model. Therefore, it should be slightly redesigned to satisfy the main requirements of a specific application, such as a rotation invariant of texture classification or robustness to noise for disparity map generation. The effect of the proposed model for other local patterns on various applications, including face detection, face recognition, and human detection, will be investigated in the near future.

How to Read a Research Paper

Step 5:

Read the introduction of the paper and answer the following questions?

1. What is the overall purpose or goal of the research?
2. What is the background or context of the research?
3. What are previous related research that has been done?
4. What are the citation of each previous work (extremely important for you)
5. Is there a review of related work or literature in the introduction?
6. Are there any notable trends or debates in the field mentioned?
7. What is the theoretical framework or approach used in the research?
8. What is the scope and focus of the research?

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Index Terms—Completed local binary pattern (CLBP), local binary pattern (LBP), local derivative pattern (LDP), support local pattern.

I. INTRODUCTION

IN RECENT years, the local binary pattern (LBP) [1] has been investigated in many applications, such as texture classification, face detection, face recognition, medical image analysis, gender classification, human detection, and object detection [2], [3], due to its robustness to illumination changes and its simplicity of computation. LBP is a nonparametric

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T. T. Nguyen is with Image Mining Group, Institut Pasteur Korea, Seongnam 463-400, Korea (e-mail: ntthuy@ip-korea.org).

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The greatest advantage of LBP is its robustness with respect to illumination change. Many stereo methods have been

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Read the Experimental Results section of the paper and answer the

1. What are the key findings or results presented in the experimental results section?
2. Do the results address the research question or hypothesis posed in the introduction?
3. Are there any unexpected or surprising results discussed in the section?
4. Are there any statistical analyses or tests mentioned, and if so, what were the results?
5. Do the results support or contradict the prior literature discussed in the introduction?
6. Are there any figures, tables, or visuals, and do they enhance your understanding of the results?
7. Which existing methods they mentioned for comparison?
8. Which datasets they used for evaluation?
9. Which conditions they conducted for testing?

texture classification by abandoning the microstructure. Although the local structure information is discarded, LBC obtains the same performance as LBP in rotation-invariant cases. In addition, a completed local count binary (CLBC) is introduced based on LBC to compete with CLBP in [26]. LBC (or LBP) is sensitive to random noise and quantization noise

IV. EXPERIMENTAL RESULTS

To assess the strength of the proposed method, we conducted robust experiments on two applications: disparity map generation and texture classification. The publicly available Middlebury [27] image dataset and the EISATS [28] synthetic vehicle sequences are used to evaluate the effectiveness of the

NGUYEN *et al.*: SUPPORT LOCAL PATTERN AND ITS APPLICATION TO DISPARITY IMPROVEMENT AND TEXTURE CLASSIFICATION

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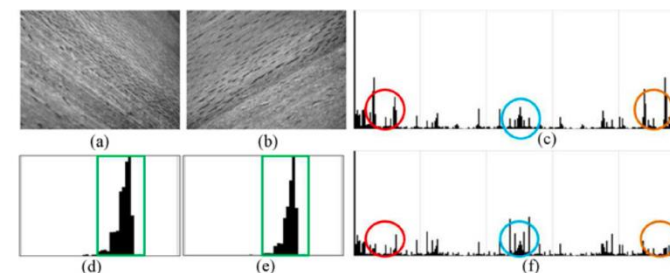


Fig. 9. (a) and (b) Two texture images. (c) and (d) SLBP_TC histograms of (a) and (b), respectively. (e) and (f) LBP histograms of (a) and (b), respectively.

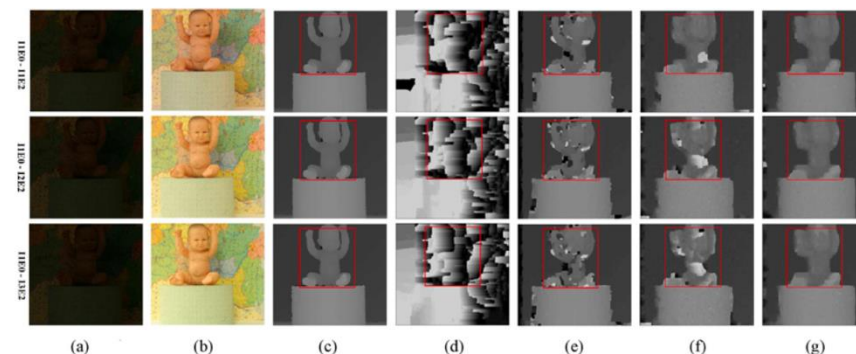


Fig. 10. Result of various stereo methods on Baby1 image pairs under various illuminations and exposures. (a) Left image. (b) Right image. (c) Ground truth. (d) AD. (e) ANCC. (f) LBP. (g) SLBP.

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CLASSIFICATION RATE (%) ON TC10 AND TC12 USING DIFFERENT SCHEMES (OUTEXT DATASET)

	$(N, R)=(8,1)$				$(N, R)=(16,2)$				$(N, R)=(24,3)$			
	TC10	TC12		Average	TC10	TC12		Average	TC10	TC12		Average
		t	h			t	h			t	h	
LTP	76.06	62.56	63.42	67.34	96.11	85.2	85.87	89.06	98.64	92.59	91.52	94.25
VAR(P, R)	90	62.93	64.35	72.42	86.71	63.47	67.26	72.48	81.66	58.98	64.18	68.80
LBP($riu2, P, R$)/VAR(P, R)	96.56	79.31	78.08	84.65	97.84	85.76	85.54	89.38	98.15	87.13	87.08	90.79
M-LBP	81.21	78.85	77.6	79.22	83.55	80.1	79.35	81.00	85.56	84.32	83.67	85.41
CLBP	96.56	90.3	92.29	93.05	98.72	93.54	93.91	95.39	98.93	95.32	94.53	96.26
SCLBP	95.45	89.45	92.88	92.59	98.05	92.77	94.21	95.01	98.85	95.25	94.45	96.18
CLBC	97.16	89.79	92.92	93.29	98.54	93.26	94.07	95.29	98.78	94.00	93.24	95.67
SCLBP_TC ($t = 0$)	97.25	89.75	92.93	93.31	98.45	93.28	93.98	95.24	99.05	93.95	93.27	95.42
SCLBP_TC ($t = 1$)	97.45	90.25	93.02	93.57	99.68	94.25	94.75	95.89	99.25	95.77	99.45	96.82
SCLBP_TC($t = 3$)	98.25	92.45	94.25	94.98	99.36	95.05	96.47	96.96	99.45	96.68	98.25	98.13
SCLBP_TC ($t = 5$)	96.45	89.25	92.15	92.62	97.89	94.02	93.25	93.05	98.26	94.86	97.22	96.78

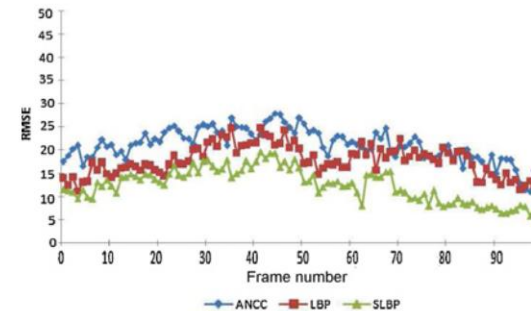
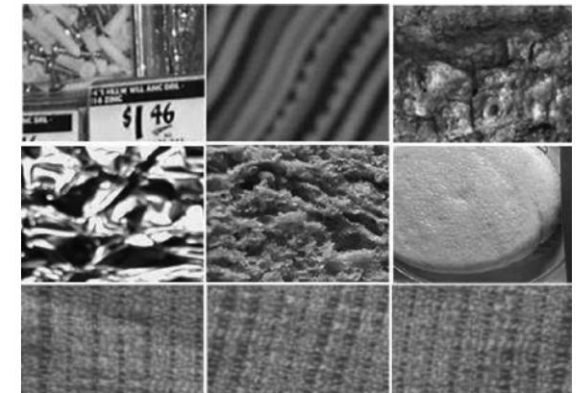


Fig. 14. RMS error of various stereo methods using the synthetic sequence with Gaussian noise.



How to Read a Research Paper

Step 6:

Read the Experimental Results section of the paper and answer the

1. What are the key findings or results presented in the experimental results section?
2. Do the results address the research question or hypothesis posed in the introduction?
3. Are there any unexpected or surprising results discussed in the section?
4. Are there any statistical analyses or tests mentioned, and if so, what were the results?
5. Do the results support or contradict the prior literature discussed in the introduction?
6. Are there any figures, tables, or visuals, and do they enhance your understanding of the results?
7. Which existing methods they mentioned for comparison?
8. Which datasets they used for evaluation?
9. Which conditions they conducted for testing?

TABLE III

CLASSIFICATION RATE (%) ON XU DATASET USING DIFFERENT SCHEMES

T	$(N, R)=(8,1)$					$(N, R)=(16,2)$					$(N, R)=(24,3)$				
	20	15	10	5	Average	20	15	10	5	Average	20	15	10	5	Average
LTP	51.40	50.08	45.06	37.70	46.06	77.45	75.66	74.28	70.88	74.57	83.25	80.58	79.46	75.55	79.71
M-LBP	83.55	81.46	79.42	69.15	78.39	85.56	82.23	79.47	70.55	79.45	87.43	85.66	84.05	82.78	84.98
CLBP	86.80	82.40	78.93	71.20	78.83	87.00	85.02	80.53	73.60	81.54	90.25	88.59	87.34	85.94	88.03
SCLBP	86.25	82.21	77.68	70.05	79.05	86.55	84.33	79.21	72.11	80.55	88.75	87.56	86.59	85.02	86.98
CLBC	87.25	83.11	79.43	72.66	80.61	87.69	86.98	82.46	74.55	82.92	91.25	89.66	87.55	86.18	88.66
SCLBP_TC ($t=0$)	87.28	83.45	79.55	73.00	80.82	87.25	87.05	82.59	75.04	82.98	92.02	89.05	88.25	85.59	88.73
SCLBP_TC ($t=1$)	88.35	83.88	81.79	73.03	81.76	87.99	86.24	82.68	74.47	83.1	93.97	92.25	91.48	87.94	91.41
SCLBP_TC ($t=3$)	87.55	82.58	80.24	71.25	80.41	86.59	86.25	81.97	74.48	83.1	82.45	93.05	90.45	86.02	88.89
SCLBP_TC ($t=5$)	86.16	82.09	79.98	70.06	79.92	85.67	84.86	80.61	73.39	81.23	92.55	89.78	89.25	85.24	89.21

TABLE IV

CLASSIFICATION RATE (%) ON KTH-TIPS DATASET USING DIFFERENT SCHEMES

T	$(N, R)=(8,1)$					$(N, R)=(16,2)$					$(N, R)=(24,3)$				
	30	20	10	5	Average	30	20	10	5	Average	30	20	10	5	Average
LTP	72.06	66.8	60.73	54.44	63.50	85.33	81.64	77.56	75.23	79.94	90.65	89.24	87.63	84.33	87.96
M-LBP	88.25	87.18	80.27	75.32	82.75	92.44	90.55	84.88	75.46	85.83	94.56	92.35	88.26	85.88	90.26
CLBP	91.91	91.59	83.97	77.46	86.23	96.32	92.41	84.85	78.45	88.01	98.35	97.64	91.55	88.48	94.00
SCLBP	90.84	89.98	81.57	77.02	84.85	96.01	91.23	83.33	77.98	87.14	96.55	95.68	89.34	87.11	92.17
CLBC	92.08	92.49	85.25	78.53	87.09	96.66	93.54	86.25	79.04	88.87	98.25	98.33	92.00	89.45	94.51
SCLBP_TC ($t=0$)	91.86	92.55	85.22	78.09	86.93	96.02	92.98	87.03	78.69	88.68	97.89	98.25	92.25	88.95	94.34
SCLBP_TC ($t=1$)	94.05	92.88	86.11	79.03	88.02	97.55	93.88	87.06	79.52	89.50	99.25	99.24	96.28	91.89	96.67
SCLBP_TC ($t=3$)	93.68	91.56	82.98	78.25	86.62	96.14	92.63	86.24	78.36	88.34	98.36	97.85	95.33	90.64	95.55
SCLBP_TC ($t=5$)	91.22	90.56	83.66	77.96	85.85	95.68	92.09	85.46	75.59	87.21	97.25	98.05	94.64	90.29	95.06

How to Read a Research Paper

Step 7:

Read the Proposed Method
section of the paper and
answer the following

1. What is the research problem or objective?
2. What is the overall approach or framework used in the method?
3. What data preprocessing or cleaning steps are involved?
4. What specific techniques or algorithms are applied?
5. Are there any assumptions made in the method?
6. How are the parameters and hyperparameters chosen or tuned?
7. How does the proposed method compare to existing methods?
8. How reproducible is the method?

II. LBP AND THE PROPOSED MODEL

In this section, we briefly review common local patterns such as the LBP and the CLBP and their disadvantages. To address the limitations of these patterns, the SLBP model is then introduced.

A. LBP

LBP [1] is a nonparametric method that captures the local structures of images by comparing each pixel with eight neighbor pixels ($N = 8$) in a 3×3 window ($R = 1$). Given a center pixel $z_c = (x, y)$ in the image, the LBP code is generated as

$$LBP_{N,R}(z_c) = \sum_{n=1}^N f(z_n, z_c) 2^n \quad (1)$$

$$f(z_n, z_c) = \begin{cases} 1, & I(z_n) \geq I(z_c) \\ 0, & I(z_n) < I(z_c) \end{cases}$$

where $I(z_c)$ and $I(z_n)$ are the gray values of the central pixel and the n th neighbor pixel, respectively. R is the radius of the neighborhood. Moreover, N neighborhood pixels can be located outside of the image grids. Their gray values can be calculated using bilinear interpolation, and the coordinate of z_n is determined by

$$(x_n, y_n) = \left[x_c + R \cos\left(\frac{2\pi n}{N}\right), y_c - R \sin\left(\frac{2\pi n}{N}\right) \right]. \quad (2)$$

Another type of LBP is the uniform pattern. A pattern is considered uniform if the number of bitwise transitions between

C. General Model of SLBP

Analyzing LBP from the view point of the local structures, we propose a supporting local pattern in order to establish a relationship among the neighborhood pixels, while preserving the simplicity of the original method. Given a local structure ($N = 8, R = 1$), the LBP code is a binary string that is established by measuring the magnitude value between a center pixel and its neighborhood pixels. An intuitive question has been raised as to whether a binary string could be established to every pixel in the local region. This means that every pixel might act as a center pixel in the local region.

Fig. 2(c) shows the results of the binary strings that are generated from every pixel in the local region from Fig. 2(a). Thus, the discriminative information can be obtained by considering each pixel in the local region as the center pixel. The structure of the local region (original microscopic pattern configuration) is not destroyed for two reasons: 1) Every pixel position is not moved in the local region; 2) From the view of each pixel position, a binary string is established by considering its intensity value relative to the intensity value of other neighborhoods. That is exactly what LBP does to establish a binary string.

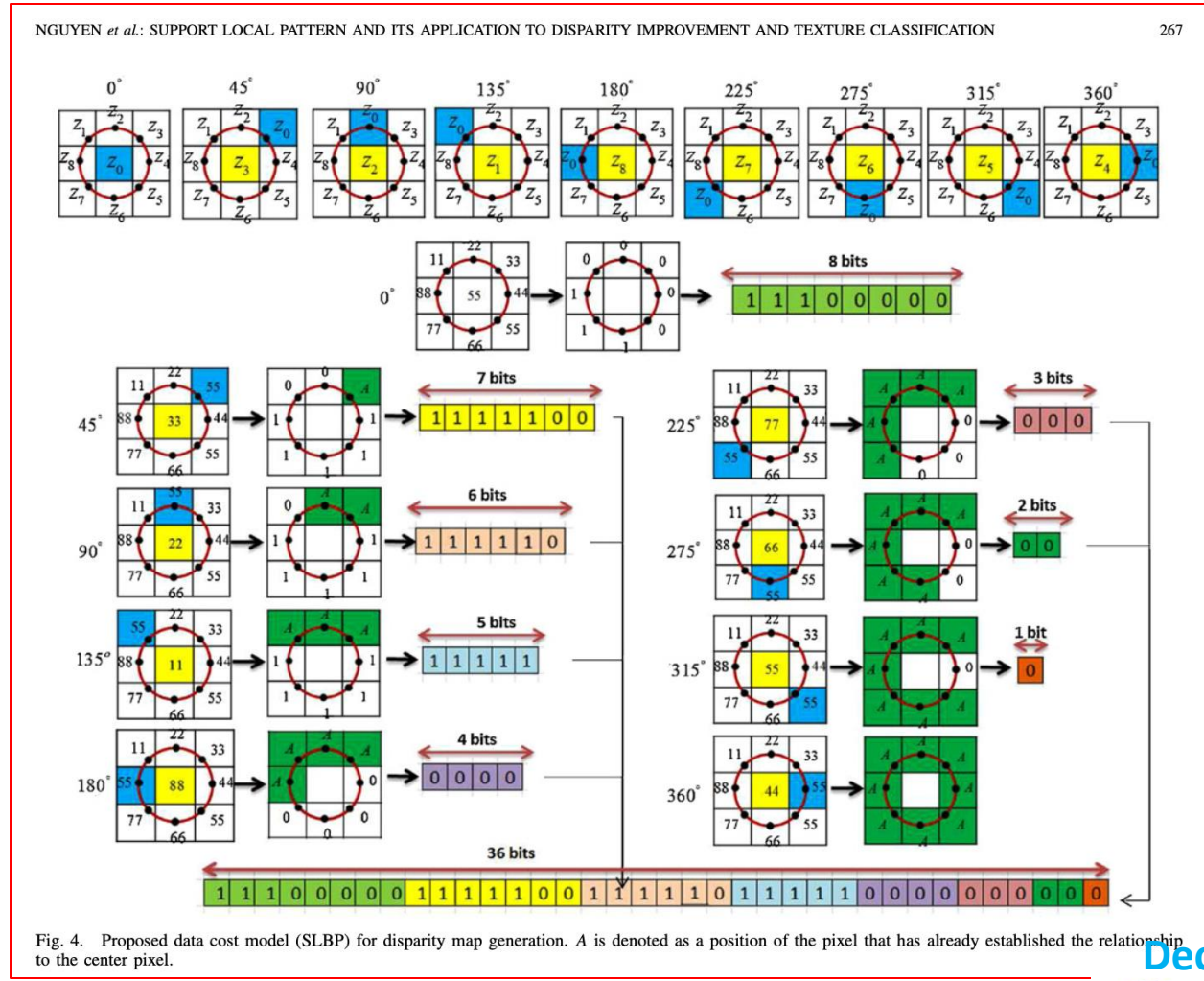
We do not consider the rotation invariant in the general SLBP model, because SLBP should be adaptively applied to specific applications based on the main characteristic of the application. For example, the disparity map generation does not require a rotation invariant, while the rotation invariant is an important property for texture classification. A simple example has been conducted to demonstrate the previous

How to Read a Research Paper

Step 7:

Read the Proposed Method section of the paper and answer the following

1. What is the research problem or objective?
2. What is the overall approach or framework used in the method?
3. What data preprocessing or cleaning steps are involved?
4. What specific techniques or algorithms are applied?
5. Are there any assumptions made in the method?
6. How are the parameters and hyperparameters chosen or tuned?
7. How does the proposed method compare to existing methods?
8. How reproducible is the method?

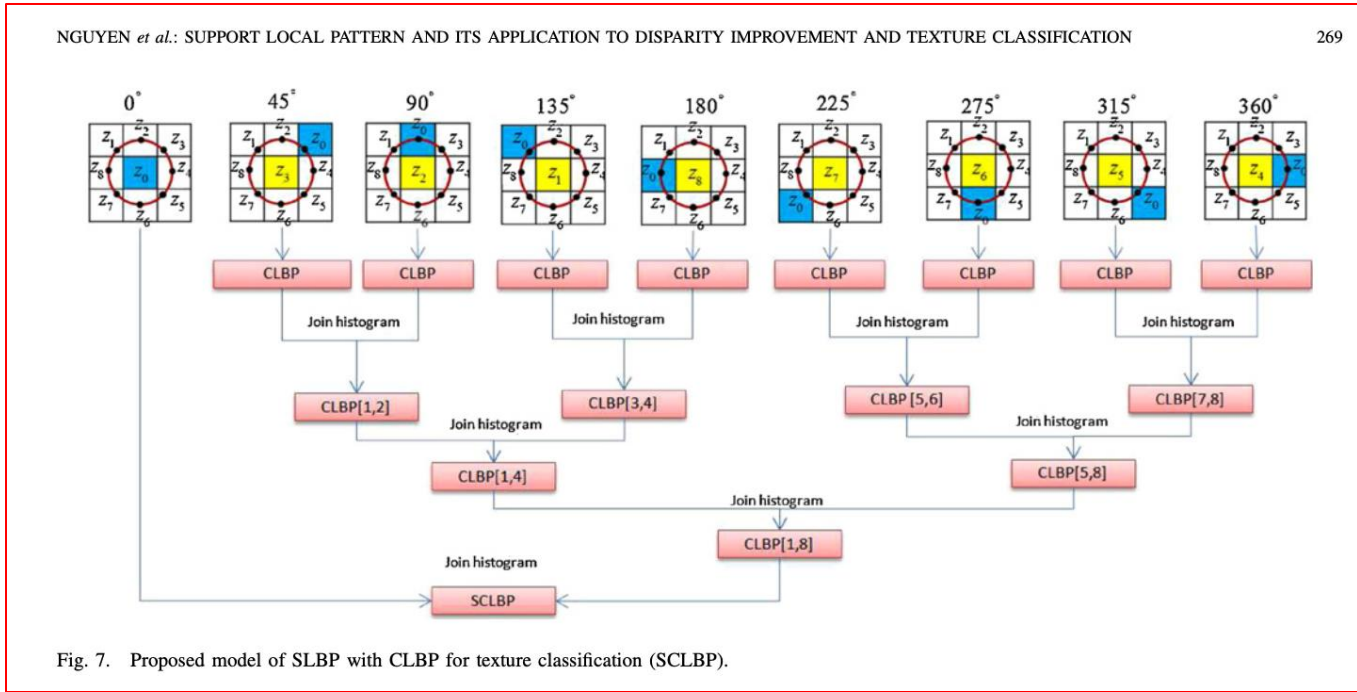


How to Read a Research Paper

Step 7:

Read the Proposed Method section of the paper and answer the following

1. What is the research problem or objective?
2. What is the overall approach or framework used in the method?
3. What data preprocessing or cleaning steps are involved?
4. What specific techniques or algorithms are applied?
5. Are there any assumptions made in the method?
6. How are the parameters and hyperparameters chosen or tuned?
7. How does the proposed method compare to existing methods?
8. How reproducible is the method?



How to Read a Research Paper

Step 8:

Read the Related Work section of the paper and answer the following question?

1. What is the purpose of the "Related Work" section?
 2. What are the key research topics or areas covered in the related work?
 3. What are the most recent and relevant studies in the field?
 4. How does the paper's research question or problem statement relate to the prior literature?
 5. Are there any gaps or limitations in the existing research that the paper aims to address?
 6. What are the main findings or insights from the related work?
 7. How does the current research build upon or extend the existing literature?

TABLE I COMPARISON OF VARIOUS STEREO METHODS FOR DISPARITY MAP GENERATION AND LBP METHODS FOR TEXTURE CLASSIFICATION AND FACE DETECTION				
Author	Algorithm	Descriptions	Type	Other
Texture classification				
Guo <i>et al.</i> [6]	Completed LBP	Incorporation with different sign and magnitude forms of local pixels.	Complementary descriptors	Use histogram combination and minimize the Chi-square distance with free training.
Liao <i>et al.</i> [7]	Dominant LBP	Combination of local information (most frequently occurring pattern) and global information (Gabor feature).	Feature selection and learning	Less sensitive to noise and histogram equalization. A support vector machine (SVM) is used.
Guo <i>et al.</i> [8]	LBP variance	Integration rotation-variant LBP histogram and global matching.	Handling rotation	Use histogram combination and minimize the chi-square distance with free training.
Zhang <i>et al.</i> [10]	Monogenic-LBP	Supplement LBP with local phase and local surface measure with Riesz transform.	Handling rotation	3-D normalized histogram and chi-square distance.
Guo <i>et al.</i> [22]	Adaptive LBP	Integration LBP with direction statistical features: mean, standard deviation, and adaptive coefficients.	Handling rotation	Chi-square distance is used to measure the histogram similarity.
Hafiane <i>et al.</i> [23]	Mean Binary Pattern	The mean value of local region is used as a threshold value.	Threshold and encoding	Capture contrast information of the local region and reduce the affect of scale changes by decomposing the images into smaller frequency ranges.
Yang and Wang [24]	LBP with Hamming distance	Integrate Hamming distance to LBP to reduce the error.	Feature selection and learning	Combine nonuniform pattern into uniform pattern by minimizing the Hamming distance.
Disparity map generation				
Kanade [12]	SAD	Measure the absolute different intensity between the left and right pixels.	Local matching	Simplest similarity measurement. Sensitive to noise.
Bolles <i>et al.</i> [13]	NCC	Find the disparity that reduce the associated error and increase the similarity.	Local matching	More complex operation than SAD. Sensitive to noise.
Zabih and Woodfill [14]	Census	Nonparametric local transform, reducing the effect of variation due to camera.	Local matching	Improved the matching result near object boundaries but still sensitive to brightness difference.
Ohta and Kanade [15]	DP	Horizontal and vertical matching with intrascanline and interscanline.	Global matching	3-D search space for long connected edges and 2-D for isolated edges. Sensitive to noise.
Boykov <i>et al.</i> [17]	Graphcut	Integration graph cut with two algorithms, α - β swap and α -expansion to compute the local minimum.	Global matching	Assume two pixels in left and right images should have the same or similar intensity. Less sensitive to noise.
Felzenszwalb and Huttenlocher [19]	BP	Maximum product algorithm is used by transmitting a message in four directions. Multiscale algorithm is used to reduce the processing time.	Global matching	Distance transform is used to minimize message cost and compute message in linear time. Less sensitive to noise.
Hirschmuller [20]	SGM-MI	Integration global methods with local stereo methods for pixel-wise matching.	Global matching	Mutual information is used for estimating matching cost. Robust to radiometric noise.
Heo <i>et al.</i> [21]	Adaptive NCC	Integration adaptive normal cross correlation with color formation model.	Global matching	Bilateral filter cost is substituted for window mean value cost. Insensitive to radiometric variation but still sensitive to multiple illumination conditions.

How to Read a Research Paper

1 Abstract

Why you should read this paper

2 Introduction

High-level overview of the algorithm

3 Related Work

What other researchers did in this field

4 Approach

Detailed description of the algorithm

5 Experiments

Training, evaluation, visualization, comparisons with other papers

6 Conclusion

Sometimes it's about next research topics

7 References

Links to all papers that ideas were used

1. Abstract

Why you should read this paper

2. Conclusion

3. Introduction

4. Experiments

5. The proposed method

6. Related Work

1. Abstract

Why you should read this paper

2. Conclusion

3. Introduction

4. Experiments

5. The proposed method

6. Related Work



Outline



➤ **How to read a research paper**

➤ **How to develop research questions**

➤ **Study and Implement Existing Methods**

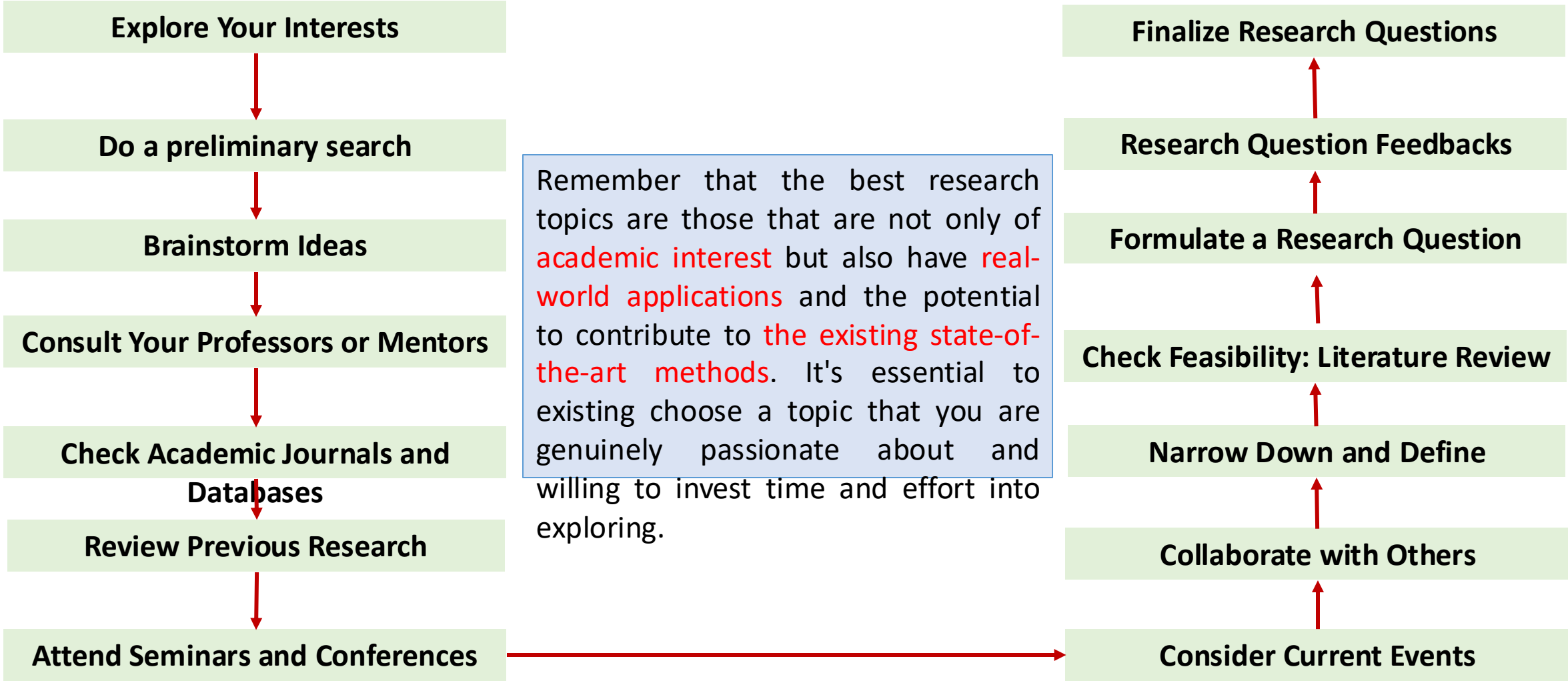
➤ **How to develop a research plan**

➤ **Implement Your Proposed Method**

➤ **Performance Evaluation**

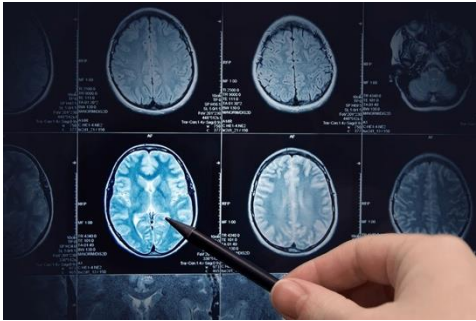
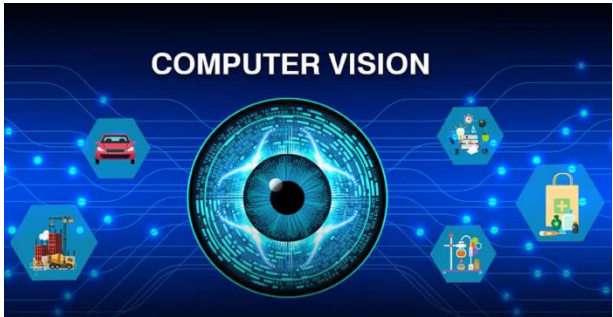
➤ **Enough Contributions for Submission**

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How to find Research Topics

Explore Your Interests



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Conduct aerial land surveys and monitor sites for security threats

WEARABLE TECH
Learns workers' habits and schedules to personalize safety alerts

3D PRINTERS
Detect and repair structural design flaws in printed construction components

AUTONOMOUS EQUIPMENT
Performs heavy-duty tasks that are dangerous or unreachable for humans

How to find Research Topics

Do a preliminary search

Read books, articles, and blogs related to your field or general interests. This will help you identify gaps in current knowledge, controversial issues, or emerging trends.

A Survey on Deep Learning Techniques for Stereo-Based Depth Estimation

Publisher: IEEE [Cite This](#) [PDF](#)

Hamid Laga ; Laurent Valentin Jospin ; Farid Boussaid ; Mohammed Bennamoun **All Authors**

56

Cites in Papers

5887

Full Text Views

Abstract

Document Sections

1 Introduction

2 Scope and Taxonomy

3 Database

Abstract:

Estimating depth from RGB images is a long-standing ill-posed problem, which has been explored for decades by the computer vision, graphics, and machine learning communities. Among the existing techniques, stereo matching remains one of the most widely used in the literature due to its strong connection to the human binocular system. Traditionally, stereo-based depth estimation has been addressed through matching hand-crafted features across multiple images. Despite the extensive amount of research, these traditional techniques still suffer in the presence of highly textured areas, large uniform regions, and occlusions. Motivated by their growing success in solving various

A Review of Vehicle Detection Techniques for Intelligent Vehicles

Publisher: IEEE [Cite This](#) [PDF](#)

Zhangyu Wang ; Jun Zhan ; Chunguang Duan ; Xin Guan ; Pingping Lu ; Kai Yang **All Authors**

28

Cites in Papers

2963

Full Text Views

Abstract

Document Sections

I. Introduction

II. Vehicle Detection Method Based on Machine

Abstract:

Robust and efficient vehicle detection is an important task of environment perception of intelligent vehicles, which directly affects the behavior decision-making and motion planning of intelligent vehicles. Due to the rapid development of sensor and computer technology, the algorithm and technology of vehicle detection have been updated rapidly. But, there are few reviews on vehicle detection of intelligent vehicles, especially covering all kinds of sensors and algorithms in recent years. This article presents a comprehensive review of vehicle detection approaches and their applications in intelligent vehicle systems to analyze the development of vehicle detection, with a specific focus on sensor types and algorithm classification. First, more than 300 research

Deep Learning for Diabetes: A Systematic Review

Publisher: IEEE [Cite This](#) [PDF](#)

Taiyu Zhu ; Kezhi Li ; Pau Herrero ; Pantelis Georgiou **All Authors**

70

Cites in Papers

4228

Full Text Views

Abstract

Document Sections

I. Introduction

II. Deep Learning Overview

III. Methodology

Abstract:

Diabetes is a chronic metabolic disorder that affects an estimated 463 million people worldwide. Aiming to improve the treatment of people with diabetes, digital health has been widely adopted in recent years and generated a huge amount of data that could be used for further management of this chronic disease. Taking advantage of this, approaches that use artificial intelligence and specifically deep learning, an emerging type of machine learning, have been widely adopted with promising results. In this paper, we present a comprehensive review of the applications of deep learning within the field of diabetes. We conducted a systematic literature search and identified three main areas that use this approach: diagnosis of diabetes, glucose management, and diagnosis of

Deep Generative Models for Synthetic Data: A Survey

Publisher: IEEE [Cite This](#) [PDF](#)

Peter Eigenschink ; Thomas Reutterer ; Stefan Vamosi ; Ralf Vamosi ; Chang Sun ; Klau... **All Authors**

1

Cites in Paper

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Abstract

Document Sections

I. Introduction

Abstract:

A growing interest in synthetic data has stimulated the development and advancement of a large variety of deep generative models for a wide range of applications. However, as this research has progressed, its streams have become more specialized and disconnected from one another. This is why models for synthesizing text data for natural language processing cannot readily be compared to models for synthesizing health

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Read books, articles, and blogs related to your field or general interests. This will help you identify gaps in current knowledge, controversial issues, or emerging trends.

Andrew Ng

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Dr. Andrew Ng is a globally recognized leader in AI (Artificial Intelligence). He is Founder of [DeepLearning.AI](#), Founder & CEO of [Landing AI](#), General Partner at [AI Fund](#), Chairman and Co-Founder of [Coursera](#) and an Adjunct Professor at Stanford University's Computer Science Department.

As a pioneer in machine learning and online education, Dr. Ng has changed countless lives through his work in AI, and has authored or co-authored over 200 research papers in machine learning, robotics and related fields. In 2013, he was named to the Time 100 list of the most influential persons in the world.

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FACULTY

Yann LeCun

• Professor

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Research Interests
AI, machine learning, computer vision, robotics, image compression



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/ Research / Research Team / Geoffrey Hinton



Geoffrey Hinton

Chief Scientific Advisor

[Website](#)

Emeritus Distinguished Professor of Computer Science, University of Toronto

Companion of the Order of Canada

Recipient ACM A.M. Turing Award

How to find Research Topics

Brainstorm Ideas

Take some time to brainstorm ideas. Write down any questions or topics that come to mind. Don't censor yourself at this stage. **Quantity over**

quality is key



Stereoreo Matching

Object Detection

Object Tracking

Text-Image Retrieval

Generative AI

Graph Neural Network

Reinforcement learning

...

How to find Research Topics

Consult Your Professors or Mentors

Ask help from your professors, mentors, or advisors. They can provide valuable insights, suggest potential research areas, or help you refine your ideas.



Stereoreo Matching

Object Detection

Object Tracking

Text-Image Retrieval

Generative AI

Graph Neural Network

Reinforcement learning

...

How to find Research Topics

Check Academic Journals and Databases

Explore academic journals, databases (like IEEE, Springer or Google Scholar), and conference proceedings in your field. This will help you discover recent research and identify areas where further study is needed.



Journals & Magazines > IEEE Transactions on Pattern ... > Volume: 44 Issue: 9

On the Confidence of Stereo Matching in a Deep-Learning Era: A Quantitative Evaluation

Publisher: IEEE [Cite This](#) [PDF](#)

Matteo Poggi; Seungryong Kim; Fabio Tosi; Sunok Kim; Filippo Aleotti; Dongbo ... [All Authors](#)

10 Cites in Papers 837 Full Text Views

Abstract

Document Sections

Abstract:

Stereo matching is one of the most popular techniques to estimate dense depth maps by finding the disparity between matching pixels on two, synchronized and rectified images. Alongside with the development of more accurate algorithms, the research community

IEEE Transactions on Pattern Analysis and Machine Intelligence



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[HTML] Review of stereo matching algorithms based on deep learning

K Zhou, X Meng, B Cheng - Computational intelligence and ..., 2020 - hindawi.com

... Stereo vision is a flourishing field, attracting the attention of ... the development of deep learning, stereo matching algorithms ... different stereo matching algorithms based on deep learning. ...

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Efficient deep learning for stereo matching

W Luo, AG Schwing, R Urtasun - Proceedings of the IEEE ..., 2016 - cv-foundation.org

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Look for gaps or unresolved issues in existing research. You can build upon or challenge previous findings to create a novel research project



Digging Into Uncertainty-Based Pseudo-Label for Robust Stereo Matching

Publisher: IEEE

Cite This

PDF

Zhelun Shen ; Xibin Song ; Yuchao Dai ; Dingfu Zhou ; Zhibo Rao ; Liangjun Zhan... [All Authors](#)

1
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Paper

60
Full
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Abstract

Document
Sections

- I. Introduction
- II. Related Work
- III. Our Approach

Abstract:

Due to the domain differences and unbalanced disparity distribution across multiple datasets, current stereo matching approaches are commonly limited to a specific dataset and generalize poorly to others. Such domain shift issue is usually addressed by substantial adaptation on costly target-domain ground-truth data, which cannot be easily obtained in practical settings. In this paper, we propose to dig into uncertainty estimation for robust stereo matching. Specifically, to balance the disparity distribution, we employ a pixel-level uncertainty-aware loss to constrain the disparity space, in this way, the unlikely

IEEE Transactions on Pattern Analysis and Machine Intelligence

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Review Previous Research

Look for gaps or unresolved issues in existing research. You can build upon or challenge previous findings to create a novel research project



Rethinking Training Strategy in Stereo Matching

Publisher: IEEE

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PDF

Zhibo Rao ; Yuchao Dai ; Zhelun Shen ; Renjie He All Authors

16
Cites in
Papers

763
Full
Text Views



Abstract

- Document Sections
- I. Introduction
 - II. Related Work
 - III. Methodology
 - IV. Experiments

Abstract:

In stereo matching, various learning-based approaches have shown impressive performance in solving traditional difficulties on multiple datasets. While most progress is obtained on a specific dataset with a dataset-specific network design, the performance on the single dataset and cross dataset affected by training strategy is often ignored. In this article, we analyze the relationship between different training strategies and performance by retraining some representative state-of-the-art methods (e.g., geometry and context network (GC-Net), pyramid stereo matching network (PSM-Net), and guided aggregation network (GA-Net), etc.). According to our research, it is surprising that the performance of these methods is significantly improved by pre-training and domain adaptation. This paper is published in **IEEE Transactions on Pattern Analysis and Machine Intelligence**. Based on

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How to find Research Topics

Collaborate with Others

Engage in discussions with peers and colleagues. They may have interesting ideas or be willing to collaborate on a research project.



How to find Research Topics

Narrow Down and Define

After brainstorming and research, narrow your list of topics to a few specific areas of interest. Ensure that the topics are feasible and have enough available resources for research.



Stereoreo Matching

Object Detection

Object Tracking

Text-Image Retrieval

Generative AI

Graph Neural Network

...

...

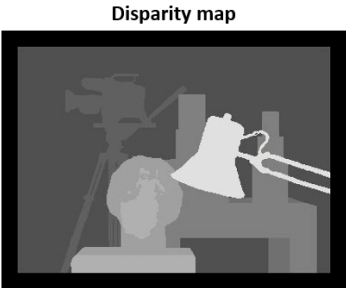
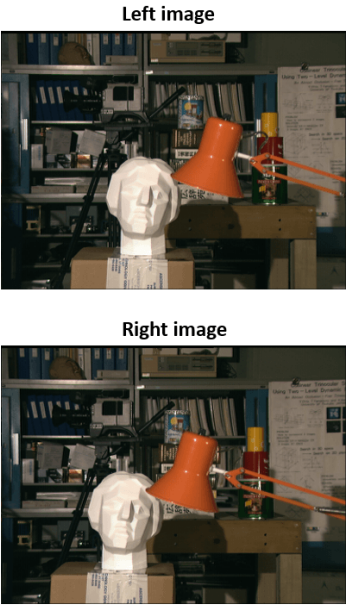
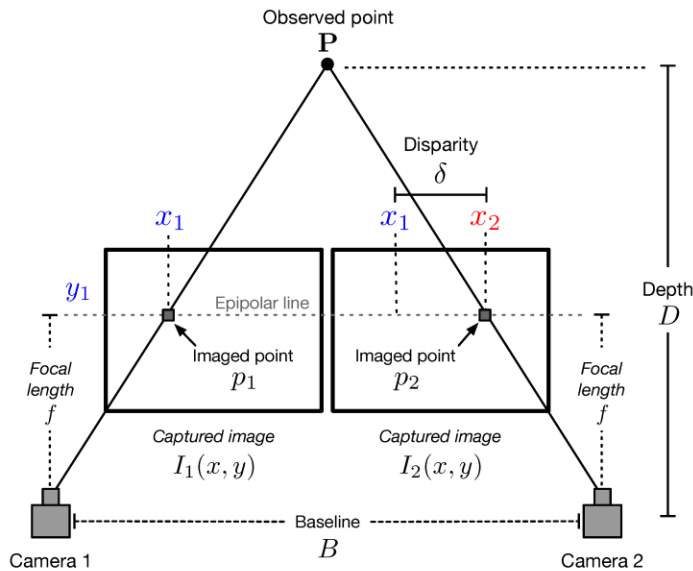
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Stereoscopic Matching



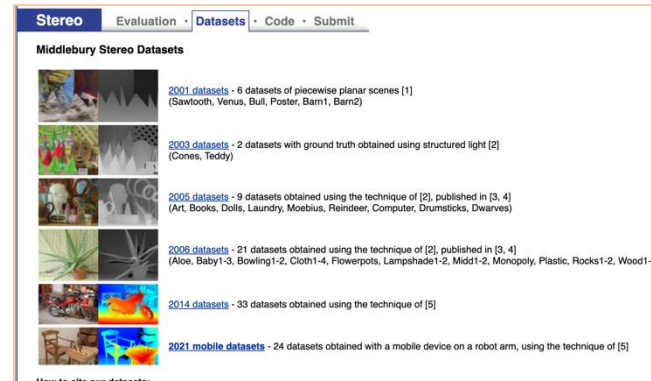
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Stereoscopic Matching



DrivingStereo: A Large-Scale Dataset for Stereo Matching in Autonomous Driving Scenarios

Publisher: IEEE

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Guorun Yang ; Xiao Song ; Chaoqin Huang ; Zhidong Deng ; Jianping Shi ; Bolei Zhou [All Authors](#)

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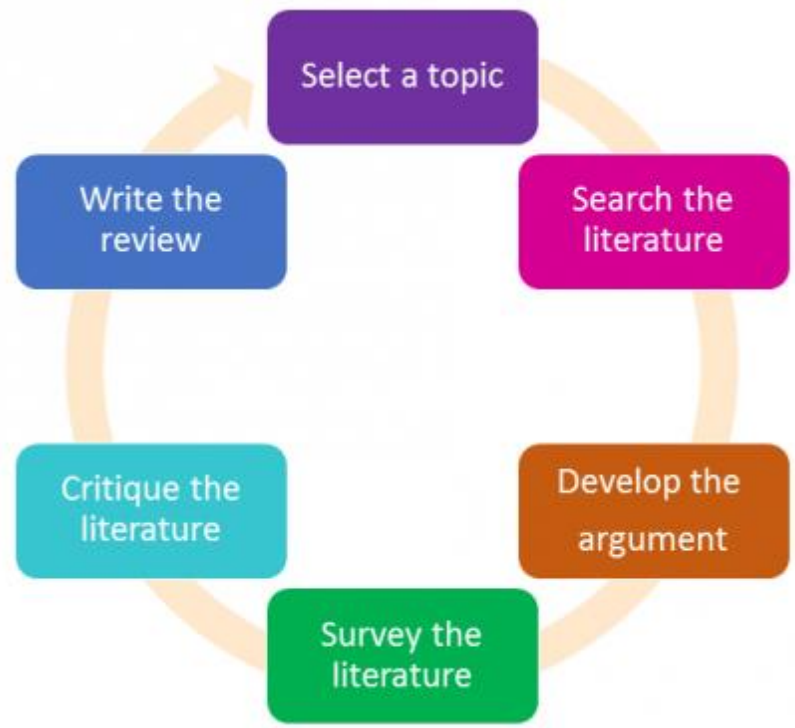


How to find Research Topics

Research the Literature

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A SURVEY ON DEEP LEARNING TECHNIQUES FOR STEREO-BASED DEPTH ESTIMATION
1

A Survey on Deep Learning Techniques for Stereo-based Depth Estimation

Hamid Laga, Laurent Valentin Jospin, Farid Boussaid, Mohammed Bennamoun *Senior Member, IEEE*

Abstract—Estimating depth from RGB images is a long-standing ill-posed problem, which has been explored for decades by the computer vision, graphics, and machine learning communities. Among the existing techniques, stereo matching remains one of the most widely used in the literature due to its strong connection to the human binocular system. Traditionally, stereo-based depth estimation has been addressed through matching hand-crafted features across multiple images. Despite the extensive amount of research, these traditional techniques still suffer in the presence of highly textured areas, large uniform regions, and occlusions. Motivated by their growing success in solving various 2D and 3D vision problems, deep learning for stereo-based depth estimation has attracted a growing interest from the community, with more than 150 papers published in this area between 2014 and 2019. This new generation of methods has demonstrated a significant leap in performance, enabling applications such as autonomous driving and augmented reality. In this article, we provide a comprehensive survey of this new and continuously growing field of research, summarize the most commonly used pipelines, and discuss their benefits and limitations. In retrospect of what has been achieved so far, we also conjecture what the future may hold for deep learning-based stereo for depth estimation research.

Index Terms—CNN, Deep Learning, 3D Reconstruction, Stereo Matching, Multi-view Stereo, Disparity Estimation, Feature Learning, Feature Matching.

1 INTRODUCTION

DEPTH estimation from one or multiple RGB images is a long standing ill-posed problem, with applications in various domains such as robotics, autonomous driving, object recognition and scene understanding, 3D modeling and animation, augmented reality, industrial control, and medical diagnosis. This problem has been extensively investigated for many decades. Among all the techniques that have been proposed in the literature, stereo matching is traditionally the most explored one due to its strong connection to the human binocular system.

The first generation of stereo-based depth estimation

previously seen objects and scenes have enabled us to build prior knowledge and develop mental models of how the 3D world looks like. The second generation of methods tries to leverage this prior knowledge by formulating the problem as a learning task. The advent of deep learning techniques in computer vision [1] coupled with the increasing availability of large training datasets, have led to a third generation of methods that are able to recover the lost dimension. Despite being recent, these methods have demonstrated exciting and promising results on various tasks related to computer vision and graphics.

In this article, we provide a comprehensive and struc-

2006.02535v1 [cs.CV] 1 Jun 2020

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TABLE 3: Taxonomy and comparison of 28 end-to-end deep learning-based disparity estimation techniques. "FCN": Fully-Connected Network, "SPN": Spatial Propagation Network. "LRCR": Left-Right Comparative Recurrent model, "MCUA": Multi-Level Context Ultra-Aggregation for Stereo Matching. "DLA": Deep layer aggregation [69], "VPP": Volumetric Pyramid Pooling. The performance is measured on KITTI2015 test dataset.

Method	Year	Feature computation		Type	Cost volume		Disparity	Refinement/post processing		Supervision	Performance	
		Architecture	Dimension		Construction	Regularization		Spatial/depth resolution	Completion/denoising		DI-all/est	DI-all/fg
FlowNetCorr [51]	2015	ConvNet	Single scale	3D	Correlation	2D ConvNet	—	Up-convolutions	Ad-hoc, variational	Supervised	—	—
DispNetC [22]	2016	ConvNet	Single scale	3D	Correlation	2D ConvNet	—	—	—	Supervised	4.34	4.32
Zhong et al. [70]	2017	ConvNet with skip conn.	Single scale	4D	Interleaved feature concat.	3D Conv, encoder-decoder	Soft argmin	Self-improvement at runtime		Self-supervised	3.57	7.12
Kendall et al. [61]	2017	ConvNet with skip conn.	Single scale	4D	Feature concat.	3D Conv, encoder-decoder, hierarchical	Soft argmax	—	—	Supervised	2.87	6.16
Pang et al. [62]	2017	ConvNet	Single scale	3D	Correlation	2D ConvNet	Upsampling+residual learning		—	Supervised	2.67	3.59
Knobelreiter et al. [71]	2017	ConvNet	Single scale	3D	Correlation	Hybrid CNN-CRF	No post-processing		—	Supervised	—	—
Chang et al. [64]	2018	SPP	Multiscale	4D	Feature concat.	3D Conv, Stacked encoder-decoders	Soft argmin	Progressive refinement		Supervised	2.32	4.62
Khamis et al. [72]	2018	ResNet	Single scale	3D	L_2	3D ConvNet	Soft argmin	Hierarchical, Upsampling+residual learning		Supervised	4.83	7.45
Liang et al. [63]	2018	ConvNet	Multiscale	3D	Correlation	2D ConvNet	Encoder-decoder	Iterative upsampling+residual learning		Supervised	2.67	3.59
Yang et al. [68]	2018	Shallow ResNet	Single scale	3D	Correlation, Left features, segmentation mask		Regression with Encoder-decoder	—	—	Self-supervised	2.25	4.07
Zhang et al. [73]	2018	ConvNet	Single scale	3D	Hand-crafted	NA	Soft argmin	Upsampling+residual learning		Self-supervised	—	—
Jie et al. [74]	2018	Constant highway net	Single scale	3D	FCN	—	RNN-based LRCR	—	—	Supervised	3.03	5.42
Ilg et al. [75]	2018	ConvNet	Single scale	3D	Correlation	Encoder-decoder, joint disparity and occlusion		Cascade of encoder-decoders, residual learning		Supervised	2.19	—
Song et al. [76]	2018	SHallow ConvNet	Single scale	3D	Correlation Encoder	Edge-guided, Context Pyramid,	Residual pyramid	—	—	Supervised	2.59	4.18
Yu et al. [77]	2018	ResNet	Single scale	3D	Feature concatenation Encoder-decoder	3D Conv + SGM	Soft argmin	—	—	Supervised	2.79	5.46
Tulyakov et al. [78]	2018	—	Single scale	4D	Compressed matching features	3D Conv	Multimodal - Sub-pixel MAP	—	—	Supervised	2.58	4.05
EMCUA et al. [65]	2019	SPP	Multiscale	4D	Feature concat.	3D Conv, MCUA scheme	Arg softmax	—	—	Supervised	2.09	4.27
Yang et al. [32]	2019	SPP	Multiscale	Pyramidal 4D	Feature difference	Conv3D blocks + Volume Pyramid Pooling	Arg softmax	Spatial & depth res. by cost volume upsampling		Supervised	2.14	3.85
Wu et al. [79]	2019	ResNet50, SPP	Multiscale	Pyramidal 4D	Feature concat.	Encoder-decoder + Feature fusion	3D Conv, soft argmin	—	—	Supervised	2.11	3.89
Yin et al. [80]	2019	DLA net	Multiscale	3D	Correlation	Density decoder	Outputs discrete matching distribution	—	—	Supervised	2.02	3.63
Chabra et al. [81]	2019	ConvNet + Vortex pooling [82]	Multiscale	3D	L_1	Dilated 3D ConvNet	Soft argmax	Upsampling+residual learning		Supervised	2.26	4.95
Duggal et al. [83]	2019	ResNet, SPP	Multiscale	3D, sparse	Correlation, Adaptive pruning with PatchMatch	Encoder-decoder	Soft argmax	Encoder	—	Supervised	2.35	3.43
Tonioni et al. [84]	2019	ConvNet	Multiscale	3D	Correlation	—	Encoder	Recursively upsampling + residual learning		Online self-adaptive	4.66	—
Yang et al. [32]	2019	ConvNet, SPP	Multiscale	Pyramid, 4D	Concatenation	Decoder, Residual blocks, VPP	Conv3D block	—	—	Supervised	2.14	3.85
Zhang et al. [85]	2019	Stacked hourglass	Single scale	4D	Concatenation	Semi-global aggregation layers, Local-guided aggregation layers	Soft argmax	—	—	Supervised	2.03	3.91
Guo et al. [86]	2019	SPP	Multiscale	Hybrid 3D-4D	Group-wise correlation	Stacked hourglass nets	Soft argmin	—	—	Supervised	2.11	3.93
Chen et al. [87]	2019	—	—	—	—	—	Single-modal weighted avg		—	Supervised	2.14	4.33
Wang et al. [88]	2019	ConvNet	Multiresolution maps	3D	L_1	Progressive refinement (3D Conv)	Soft argmin	Upsampling, residual learning	Spatial propagation network	Supervised	—	—

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Interestingly, Table 9 and Fig. 10 show that most of the methods achieve similar reconstruction accuracies. The only exception is iResNet [63] trained on Kitti2015 and on ROB [152], which had more than 90%, respectively 55%, of pixels with an error that is larger than 5 pixels. In all methods, less than 5% of the pixels had an error that is less than 2 pixels. This suggests that achieving sub-pixel accuracy remains an important challenge for future research.

Note that SegStereo [68], which is self-supervised, achieves a similar or better performance than many of the supervised methods. Also, the unsupervised self-adaptation method of Tonioni *et al.* [129], which takes the baseline DispNet-Corr1D network [22] trained on KITTI 2012 and adapts it to KITTI2015 and Middlebury 2014, achieves one of the best performances on the foreground regions.

In terms of the visual quality of the estimated disparities, see Fig. 11, we observe that most of the methods were able to recover the overall shape of trees but fail to reconstruct the details especially the leaves. The reconstruction errors are high in flat areas and around object boundaries. Also, highly reflective materials and poor lighting conditions remain a big challenge to these methods as shown in Fig. 11-(b). The supplementary material provides more results on the four stereo pairs of Fig. 9.

9 FUTURE RESEARCH DIRECTIONS

Deep learning methods for stereo-based depth estimation have achieved promising results. The topic, however, is still in its infancy and further developments are yet to be expected. In this section, we present some of the current issues and highlight directions for future research.

(1) *Camera parameters.* Most of the stereo-based techniques surveyed in this article require rectified images. Multi-view stereo techniques use Plane-Sweep Volumes or back-projected images/features. Both image rectification and

(2) *Lighting conditions and complex material properties.* Poor lighting conditions and complex materials properties remain a challenge to most of the current methods, see for example Fig. 11-(b). Combining object recognition, high-level scene understanding, and low-level feature learning can be one promising avenue to address these issues.

(3) *Spatial and depth resolution.* Most of the current techniques do not handle high resolution input images and generally produce depth maps of low spatial and depth resolution. Depth resolution is particularly limited, making the methods unable to reconstruct thin structures, *e.g.*, vegetation and hair, and structures located at a far distance from the camera. Although refinement modules can improve the resolution of the estimated depth maps, the gain is still small compared to the resolution of the input images. This has recently been addressed using hierarchical techniques, which allow on-demand reports of disparity by capping the resolution of the intermediate results [32]. In these methods, low resolution depth maps can be produced in realtime, and thus can be used on mobile platforms, while high resolution maps would require more computation time. Producing, in realtime, accurate maps of high spatial and depth resolutions remains a challenge for future research.

(4) *Realtime processing.* Most deep learning methods for disparity estimation use 3D and 4D cost volumes, which are processed and regularized using 2D and 3D convolutions. They are expensive in terms of memory requirements and processing time. Developing lightweight, and subsequently fast, end-to-end deep networks remains a challenging avenue for future research.

(5) *Disparity range.* Existing techniques uniformly discretize the disparity range. This results in multiple issues. In particular, although the reconstruction error can be small in the disparity space, it can result in an error of meters in the

How to find Research Topics

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Rethinking Training Strategy in Stereo Matching

Publisher: IEEE

[Cite This](#)

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Zhibo Rao ; Yuchao Dai ; Zhelun Shen ; Renjie He [All Authors](#)

16

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Abstract

Document
Sections

- I. Introduction
- II. Related Work
- III. Methodology
- IV. Experiments

Abstract:

In stereo matching, various learning-based approaches have shown impressive performance in solving traditional difficulties on multiple datasets. While most progress is obtained on a specific dataset with a dataset-specific network design, the performance on the single dataset and cross dataset affected by training strategy is often ignored. In this article, we analyze the relationship between different training strategies and performance by retraining some representative state-of-the-art methods (e.g., geometry and context network (GC-Net), pyramid stereo matching network (PSM-Net), and guided aggregation network (GA-Net), etc.). According to our research, it is surprising that the performance of networks on single or cross datasets is significantly improved by pre-training and data augmentation without any particular structure acquirement. Based on

How to Develop Research Question

Formulate a Research Question

Turn your topic into a clear and focused research question. A well-defined question is crucial for guiding your research.



Stereo Matching – 3D Reconstruction – Depth Estimation

Camera parameters

Lighting conditions and complex material properties.

Spatial and depth resolution

Realtime processing.

Disparity range.

Training.

Automatically learning the network architecture, its activation functions, and its parameters from data



How to Develop Research Question

Seek Feedback

Share your research question with colleagues, professors, or mentors to get their feedback and suggestions



Stereo Matching – 3D Reconstruction – Depth Estimation

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Realtime processing.

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How to Develop Research Question

Finalize Research Questions

Based on the feedback, refine your research topic and question as needed. It's common for topics to evolve during this process.



Stereo Matching – 3D Reconstruction – Depth Estimation

RQ1: How to improve the performance of existing stereo matching methods under various lighting conditions and complex material properties?

RQ2: How to obtain a real-time processing for current state-of-the-art stereo matching-based deep learning methods?

RQ3: How to design an automatically learning the network architecture, its activation functions, and its parameters from data ?

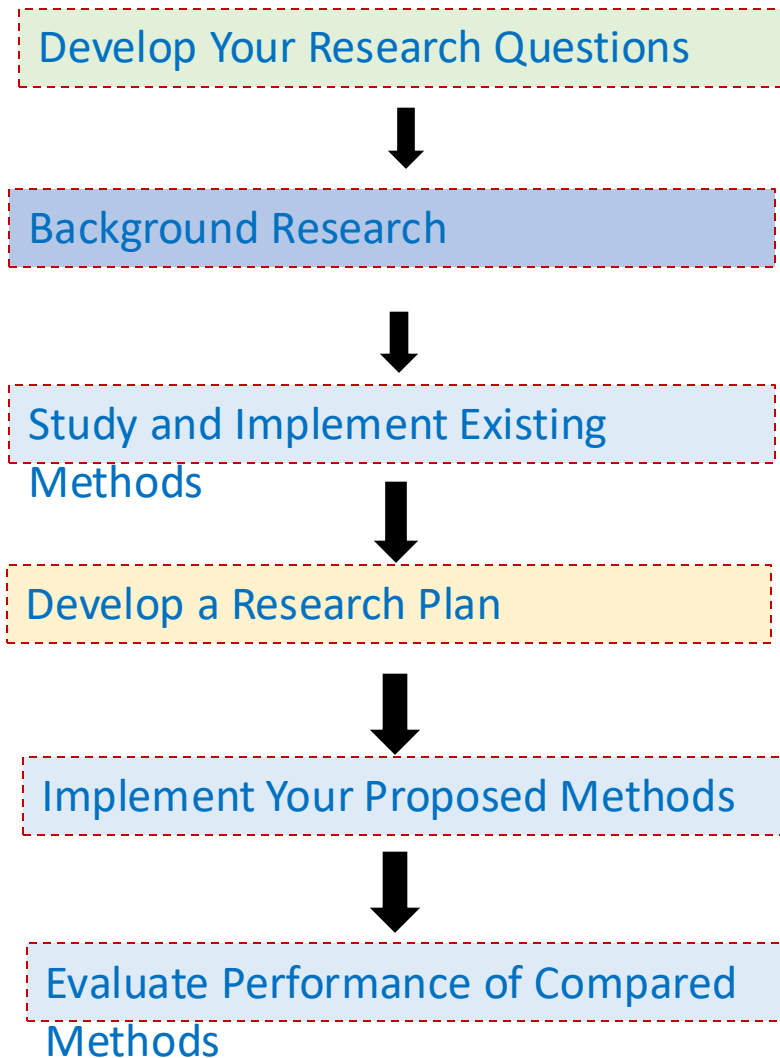
Remember that the best research topics are those that are not only of academic interest but also have real-world relevance and the potential to contribute to the existing body of knowledge. It's essential to choose a topic that you are genuinely passionate about and willing to invest time and effort into exploring.

Outline

- **How to read a research paper**
- **How to develop research questions**
- **Study and Implement Existing Methods**
- **How to develop a research plan**
- **Implement Your Proposed Method**
- **Performance Evaluation**
- **Enough Contributions for Submission**



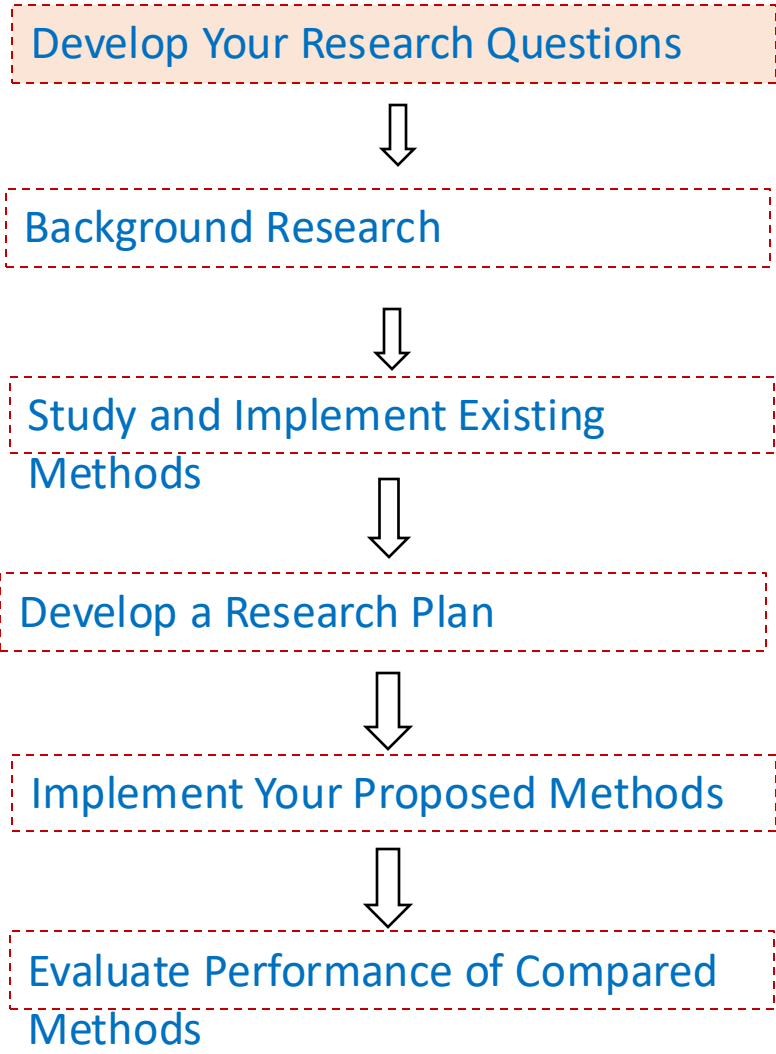
How to do Research



- 1. Research Title:**
Provide a clear and concise title that reflects the main focus of your research
- 2. Introduction**
Briefly introduce your research topic, explaining its significance and relevance.
- 3. Background and Literature Review**
Summarize the existing knowledge and research related to your topic. Highlight key studies, theories, or gaps in the literature that your research aims to address.
- 4. Research Objectives**
What do you aim to achieve or discover through your study?
- 5. Research Questions**
Formulate the research questions or hypotheses that will guide your investigation
- 6. Methodology**
Detail the research design, methods, and procedures you plan to employ
- 7. Timeline**
Provide a timeline for your research activities. Include milestones, deadlines, and key events.
- 8. Resources**
Outline the resources you'll need for your research, such as access to specific databases, equipment, or funding.
- 9. Expected Result**
Discuss the anticipated outcomes of your research.
- 10. Significant and Contribution**
Explain the potential significance of your research.

How to do Research

Stereo Matching – Depth Estimation

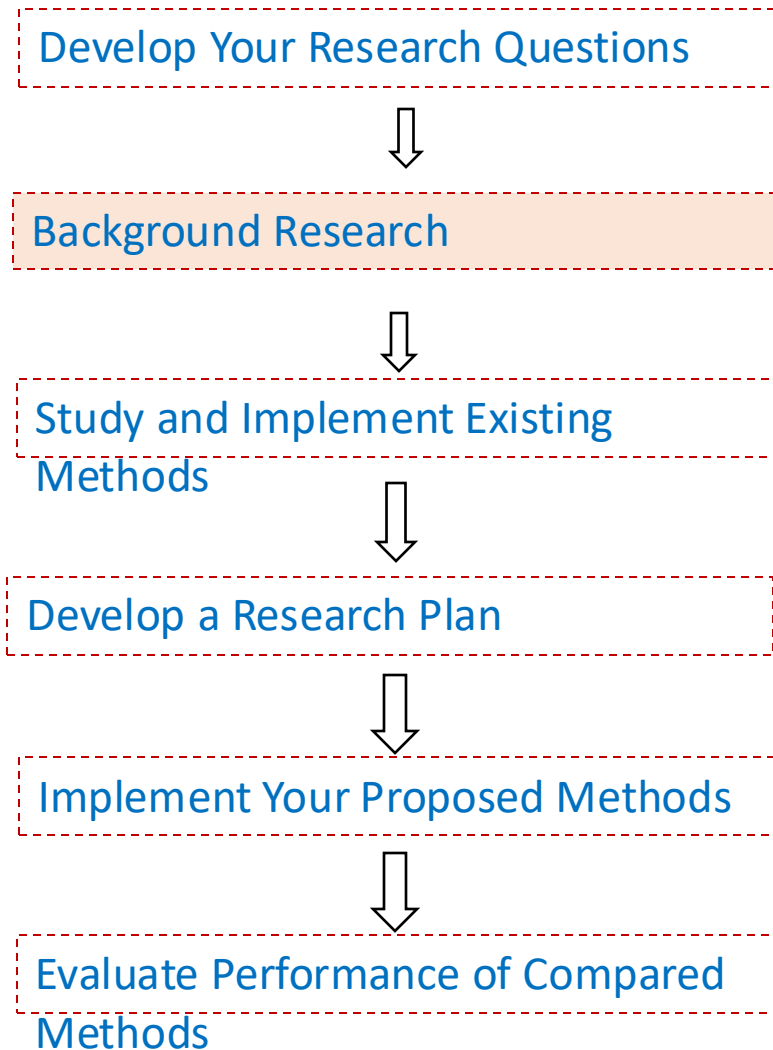


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Stereo Matching: Background



- 1. Identify Keywords and Concepts**
- 2. Conduct a Literature Review**
- 3. Review Relevant Journals and Conferences**
- 4. Identify Key Theories and Frameworks**
- 5. Explore Citations**
- 6. Check for Recent Developments**

Some popular databases include PubMed, Google Scholar, IEEE Xplore, Springer, Elsevier, and JSTOR,

Develop Your Research Questions



Background Research



Study and Implement Existing
Methods



Develop a Research Plan



Implement Your Proposed Methods



Evaluate Performance of Compared
Methods

A SURVEY ON DEEP LEARNING TECHNIQUES FOR STEREO-BASED DEPTH ESTIMATION

1

A Survey on Deep Learning Techniques for Stereo-based Depth Estimation

Hamid Laga, Laurent Valentin Jospin, Farid Boussaid, Mohammed Bennamoun *Senior Member, IEEE*

Abstract—Estimating depth from RGB images is a long-standing ill-posed problem, which has been explored for decades by the computer vision, graphics, and machine learning communities. Among the existing techniques, stereo matching remains one of the most widely used in the literature due to its strong connection to the human binocular system. Traditionally, stereo-based depth estimation has been addressed through matching hand-crafted features across multiple images. Despite the extensive amount of research, these traditional techniques still suffer in the presence of highly textured areas, large uniform regions, and occlusions. Motivated by their growing success in solving various 2D and 3D vision problems, deep learning for stereo-based depth estimation has attracted a growing interest from the community, with more than 150 papers published in this area between 2014 and 2019. This new generation of methods has demonstrated a significant leap in performance, enabling applications such as autonomous driving and augmented reality. In this article, we provide a comprehensive survey of this new and continuously growing field of research, summarize the most commonly used pipelines, and discuss their benefits and limitations. In retrospect of what has been achieved so far, we also conjecture what the future may hold for deep learning-based stereo for depth estimation research.

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previously seen objects and scenes have enabled us to build prior knowledge and develop mental models of how the 3D world looks like. The **second** generation of methods tries to leverage this prior knowledge by formulating the problem as a learning task. The advent of deep learning techniques in computer vision [1] coupled with the increasing availability of large training datasets, have led to a **third** generation of methods that are able to recover the lost dimension. Despite being recent, these methods have demonstrated exciting and promising results on various tasks related to computer vision and graphics.

Develop Your Research Questions



Background Research



Study and Implement Existing
Methods



Develop a Research Plan



Implement Your Proposed Methods



Evaluate Performance of Compared
Methods

Survey on Semantic Stereo Matching / Semantic Depth Estimation

Viny Saajan Victor¹ and Peter Neigel²

¹ victor@rhrk.uni-kl.de

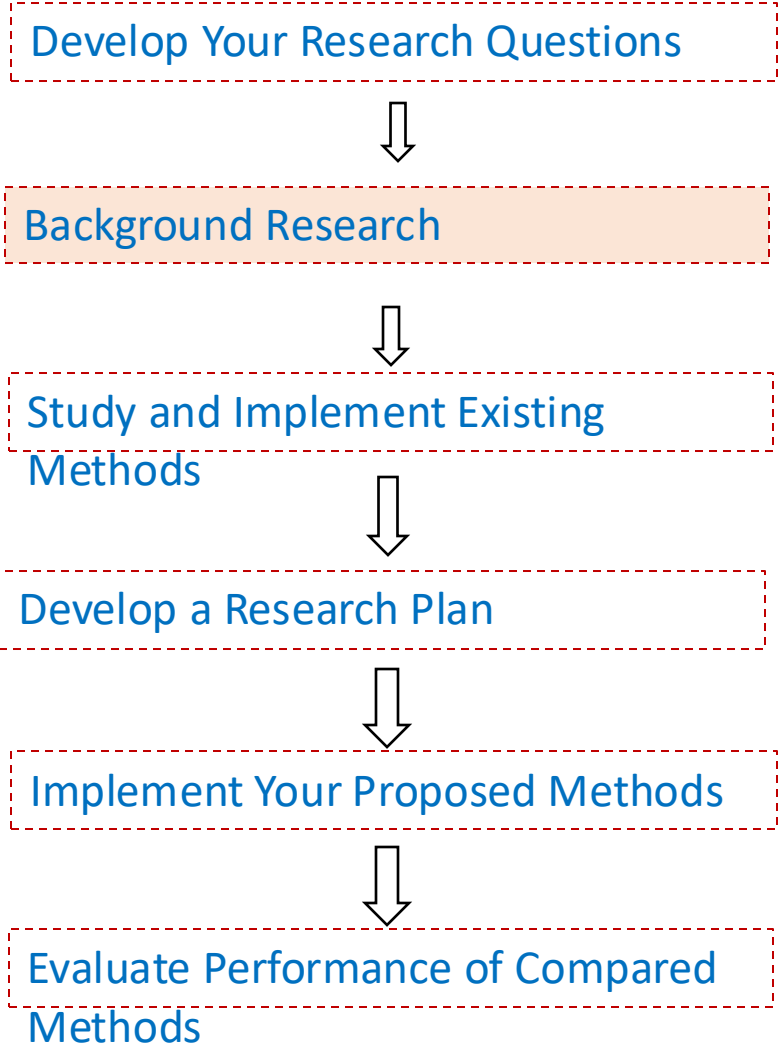
² peter.neigel@dfki.de

Abstract. Stereo matching is one of the widely used techniques for inferring depth from stereo images owing to its robustness and speed. It has become one of the major topics of research since it finds its applications in autonomous driving, robotic navigation, 3D reconstruction, and many other fields. Finding pixel correspondences in non-textured, occluded and reflective areas is the major challenge in stereo matching. Recent developments have shown that semantic cues from image segmentation can be used to improve the results of stereo matching. Many deep neural network architectures have been proposed to leverage the advantages of semantic segmentation in stereo matching. This paper aims to give a comparison among the state of art networks both in terms of accuracy and in terms of speed which are of higher importance in real-time applications.

Keywords: Stereo Matching, Semantic Segmentation, Depth Estimation, Conditional Random Fields, Warping Error, Multiscale Context Intertwining

1 Introduction

1.1 Stereo Matching: Stereo Matching is a technique used to find point correspondences between two images of a scene acquired by known cameras. The traditional approaches such as Semi-Global Matching [5] includes well-defined steps such as cost matching (calculating matching cost for each pixel in one of the stereo images depending on the corresponding pixel in the other image), cost aggregation (calculating matching cost for range of disparity values to form cost volumes), disparity estimation (calculating the disparity from cost volume by minimizing the cost function) and disparity post-processing (improving the disparity). These approaches make assumptions that the scene is uniformly illuminated and textured.



1582

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 31, NO. 9, SEPTEMBER 2009

Evaluation of Stereo Matching Costs on Images with Radiometric Differences

Heiko Hirschmüller and Daniel Scharstein, *Member, IEEE*

Abstract—Stereo correspondence methods rely on matching costs for computing the similarity of image locations. We evaluate the insensitivity of different costs for passive binocular stereo methods with respect to radiometric variations of the input images. We consider both pixel-based and window-based variants like the absolute difference, the sampling-insensitive absolute difference, and normalized cross correlation, as well as their zero-mean versions. We also consider filters like LoG, mean, and bilateral background subtraction (BilSub) and nonparametric measures like Rank, SoftRank, Census, and Ordinal. Finally, hierarchical mutual information (HMI) is considered as pixelwise cost. Using stereo data sets with ground-truth disparities taken under controlled changes of exposure and lighting, we evaluate the costs with a local, a semiglobal, and a global stereo method. We measure the performance of all costs in the presence of simulated and real radiometric differences, including exposure differences, vignetting, varying lighting, and noise. Overall, the ranking of methods across all data sets and experiments appears to be consistent. Among the best costs are BilSub, which performs consistently very well for low radiometric differences; HMI, which is slightly better as pixelwise matching cost in some cases and for strong image noise; and Census, which showed the best and most robust overall performance.

Index Terms—Stereo, matching cost, performance evaluation, radiometric differences.

1 INTRODUCTION

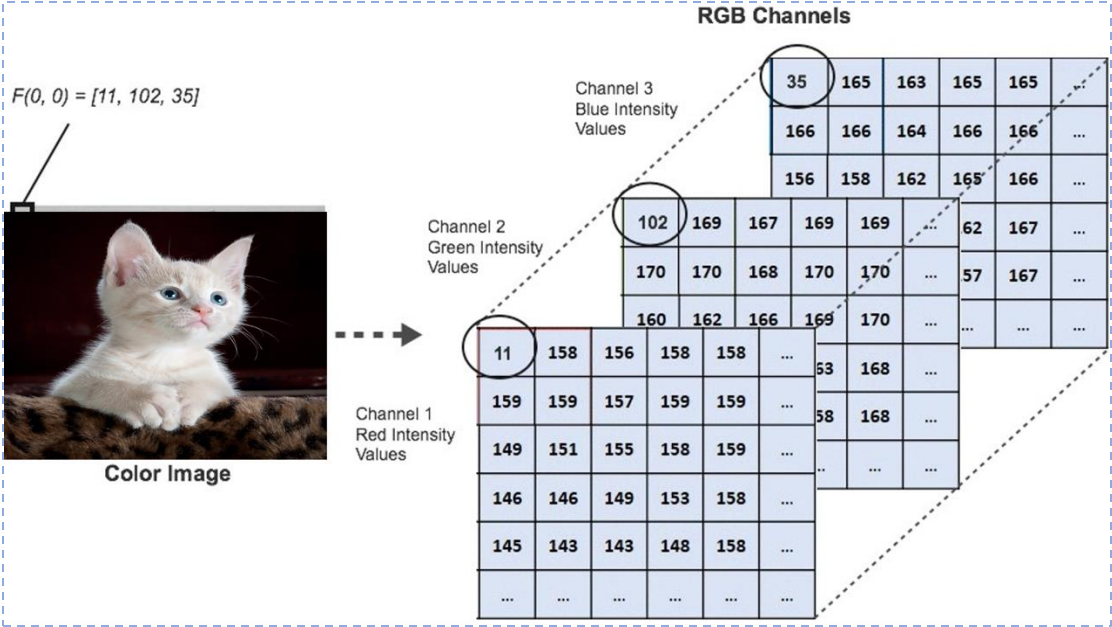
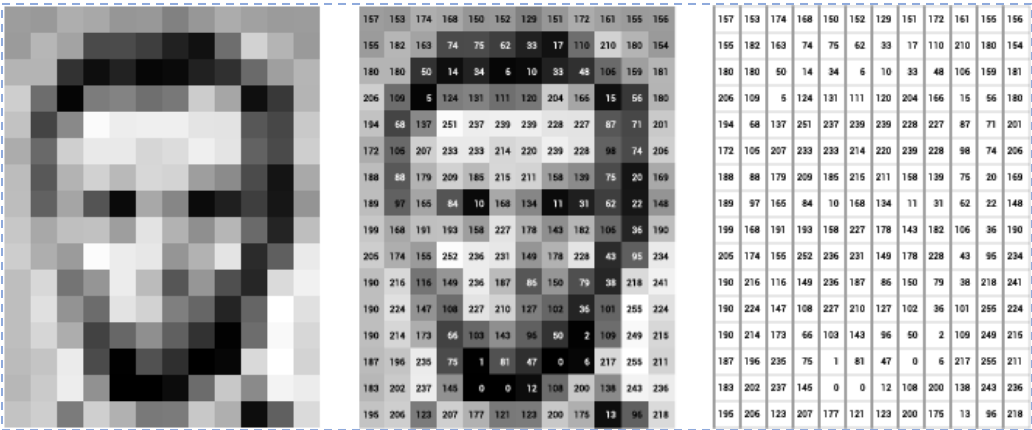
ALL passive stereo correspondence algorithms have a way of measuring the similarity of image locations. Typically, a *matching cost* is computed at each pixel for all disparities under consideration. The simplest matching costs assume constant intensities at matching image

outdoors). Similar situations arise when matching aerial or satellite images.

Due to all of the above reasons, it is safe to say that any real-world stereo application requires radiometric robustness. This includes existing commercial systems, which

Stereo Matching: Background

What is a digital image?



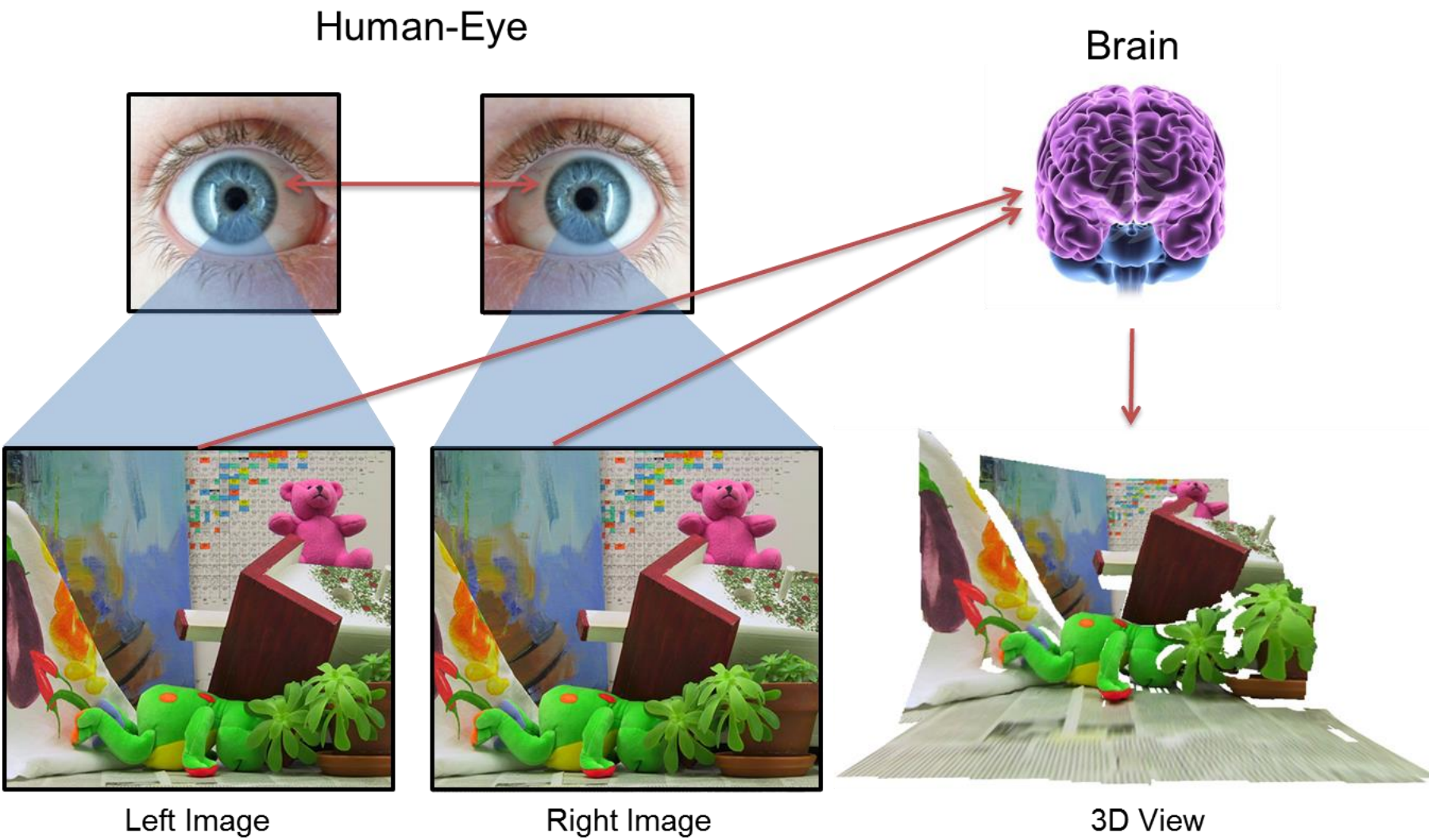
Color Image



Graycalse Image

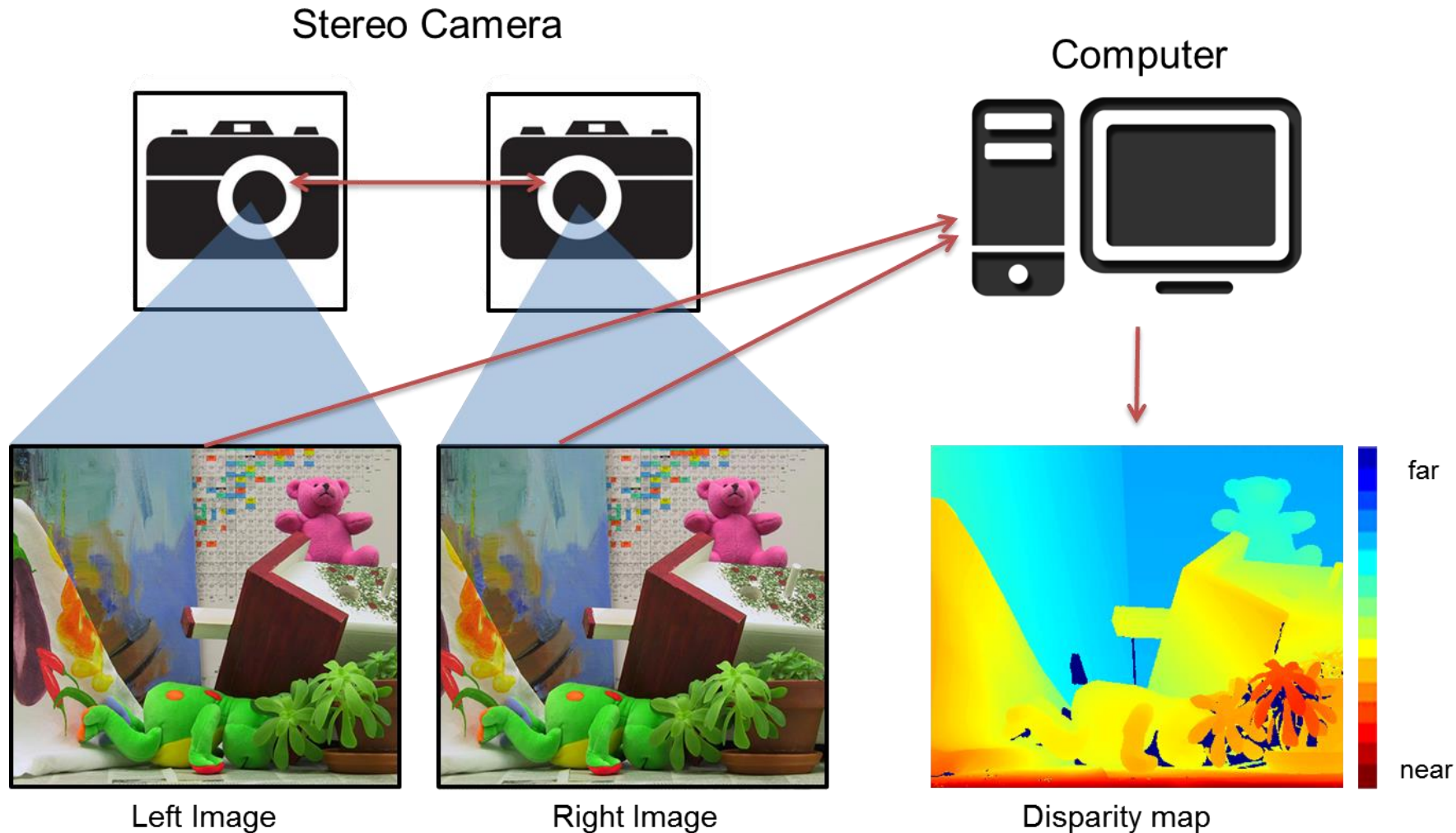
Stereo Matching: Background

Human perception for 3D information



Stereo Matching: Background

We generate 3D information using disparity maps obtained from stereo matching algorithm



Stereo Matching: Background

Challenges in stereo matching



Normal stereo pair



ADCensus



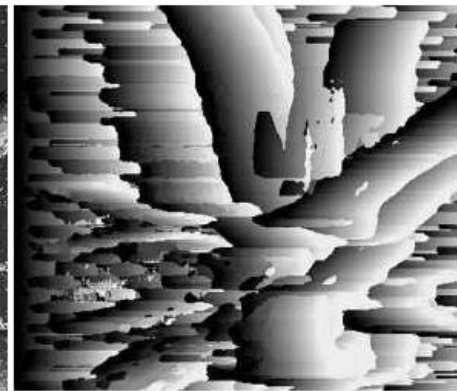
SAD



Stereo pair with different illumination

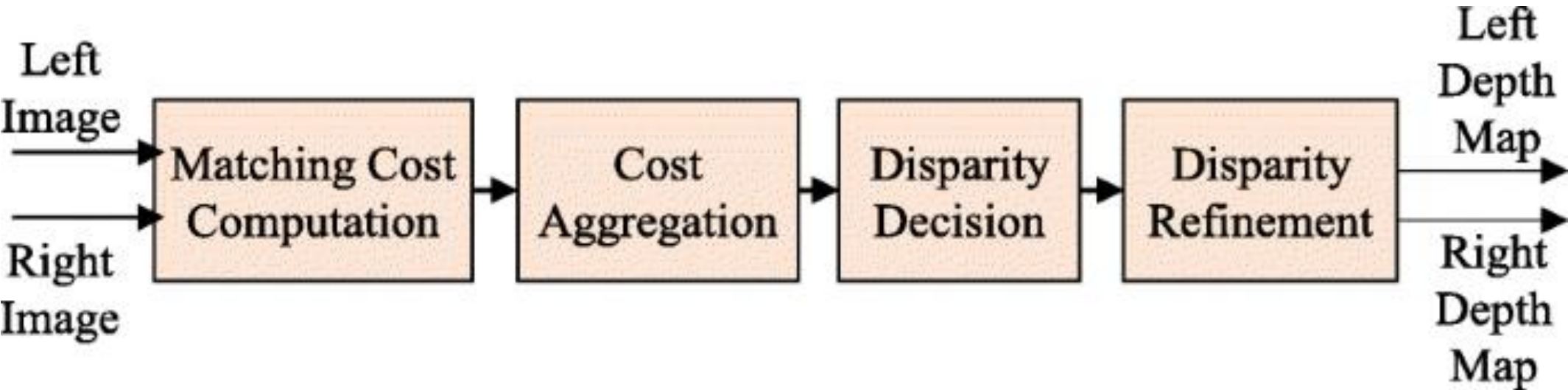


ADCensus

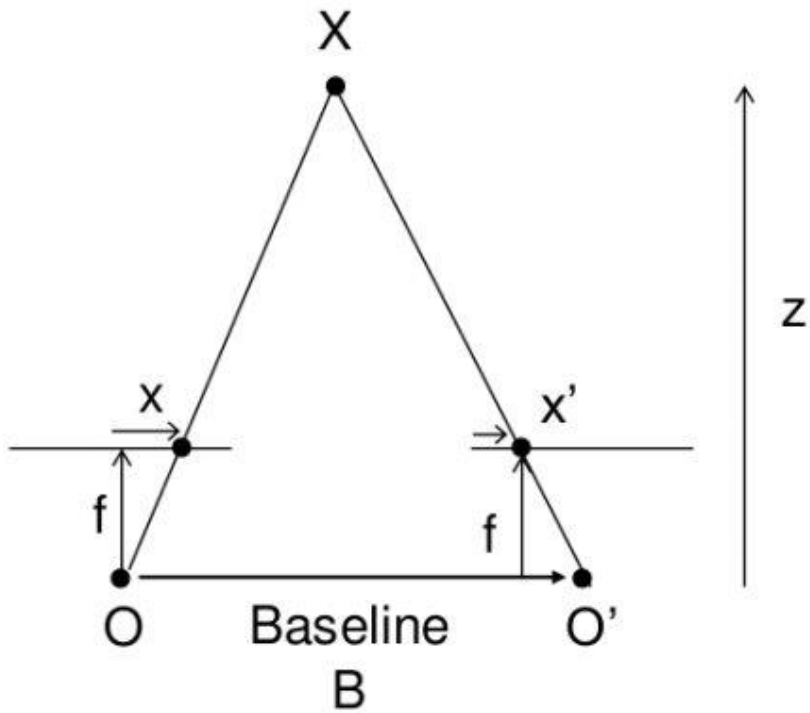


SAD

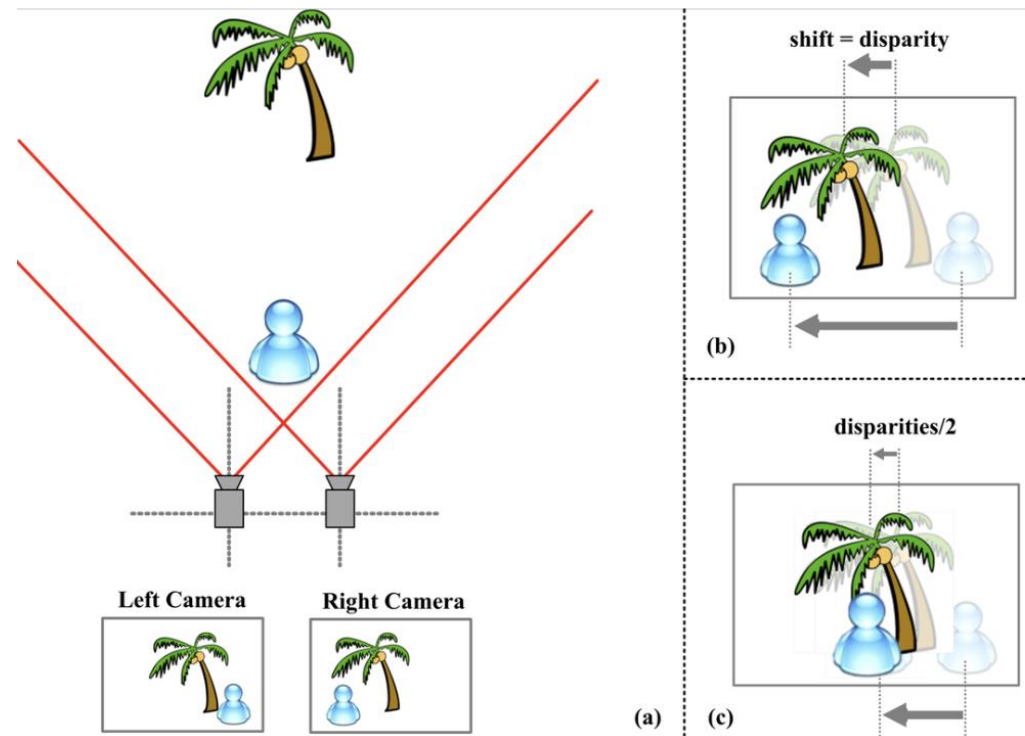
Stereo Matching: Background



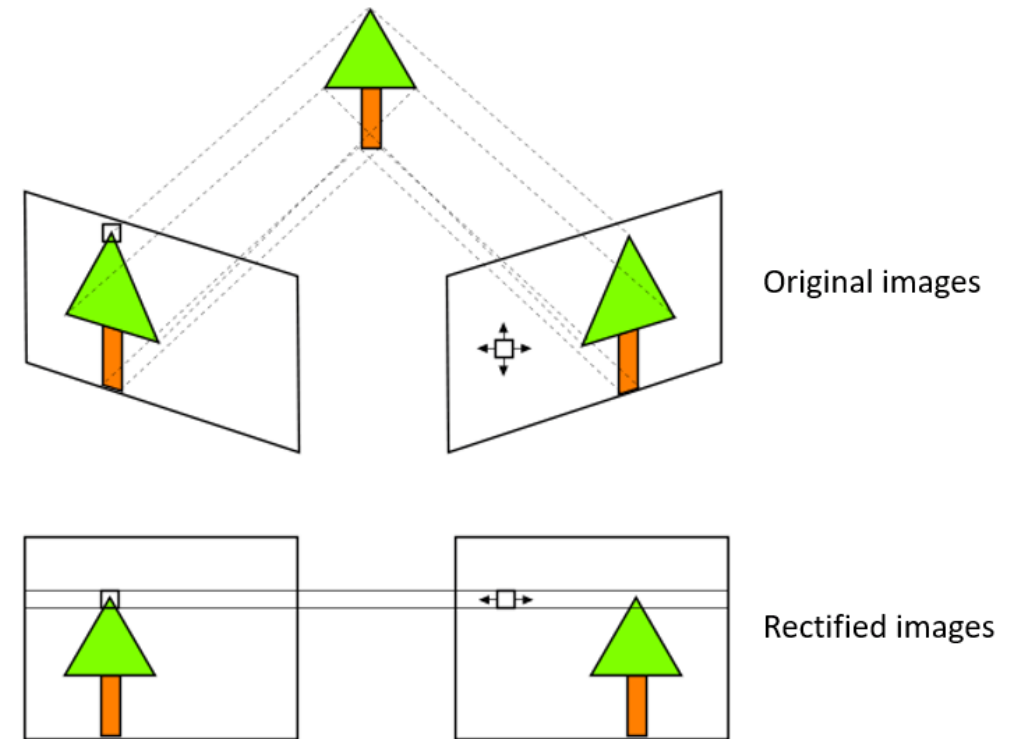
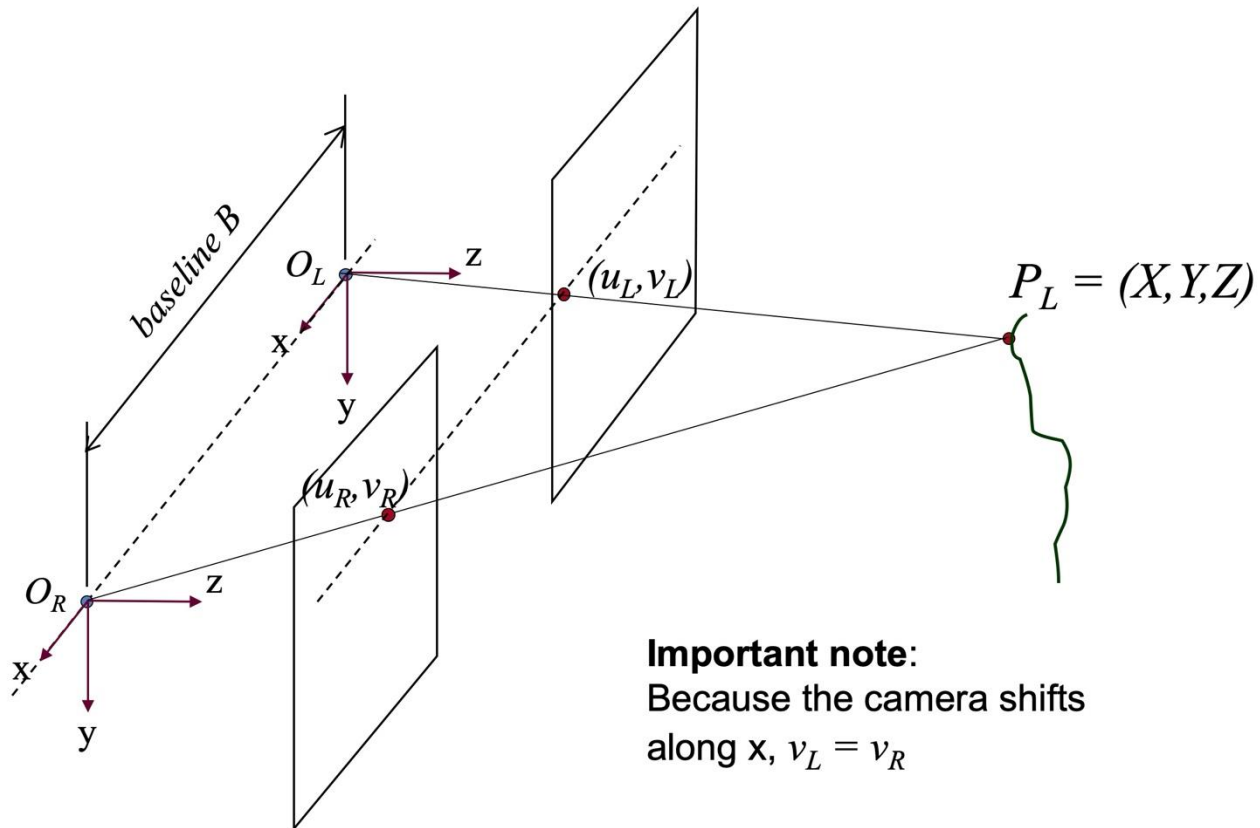
Stereo Matching: Background



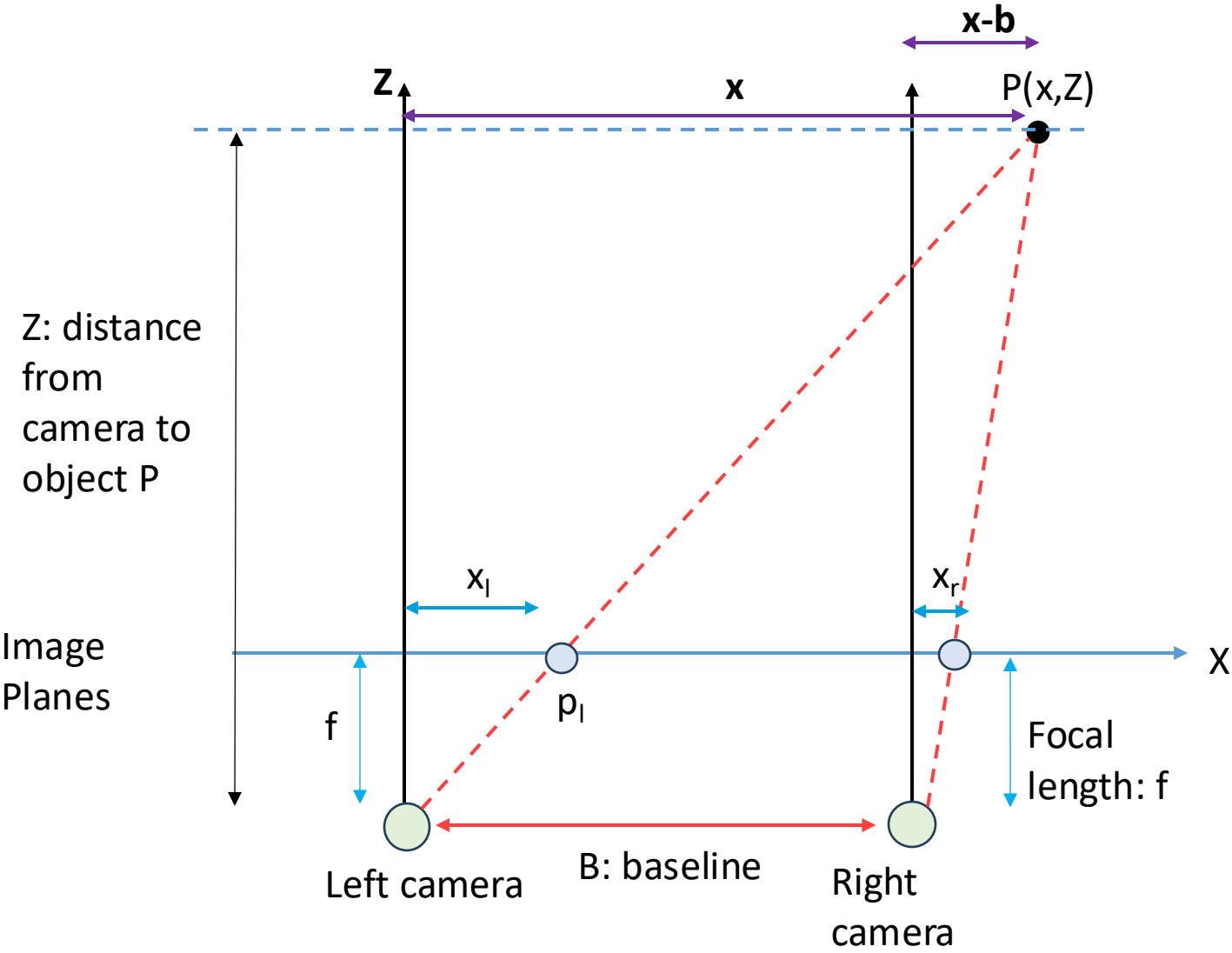
$$disparity = x - x' = \frac{Bf}{Z}$$



Stereo Matching: Background



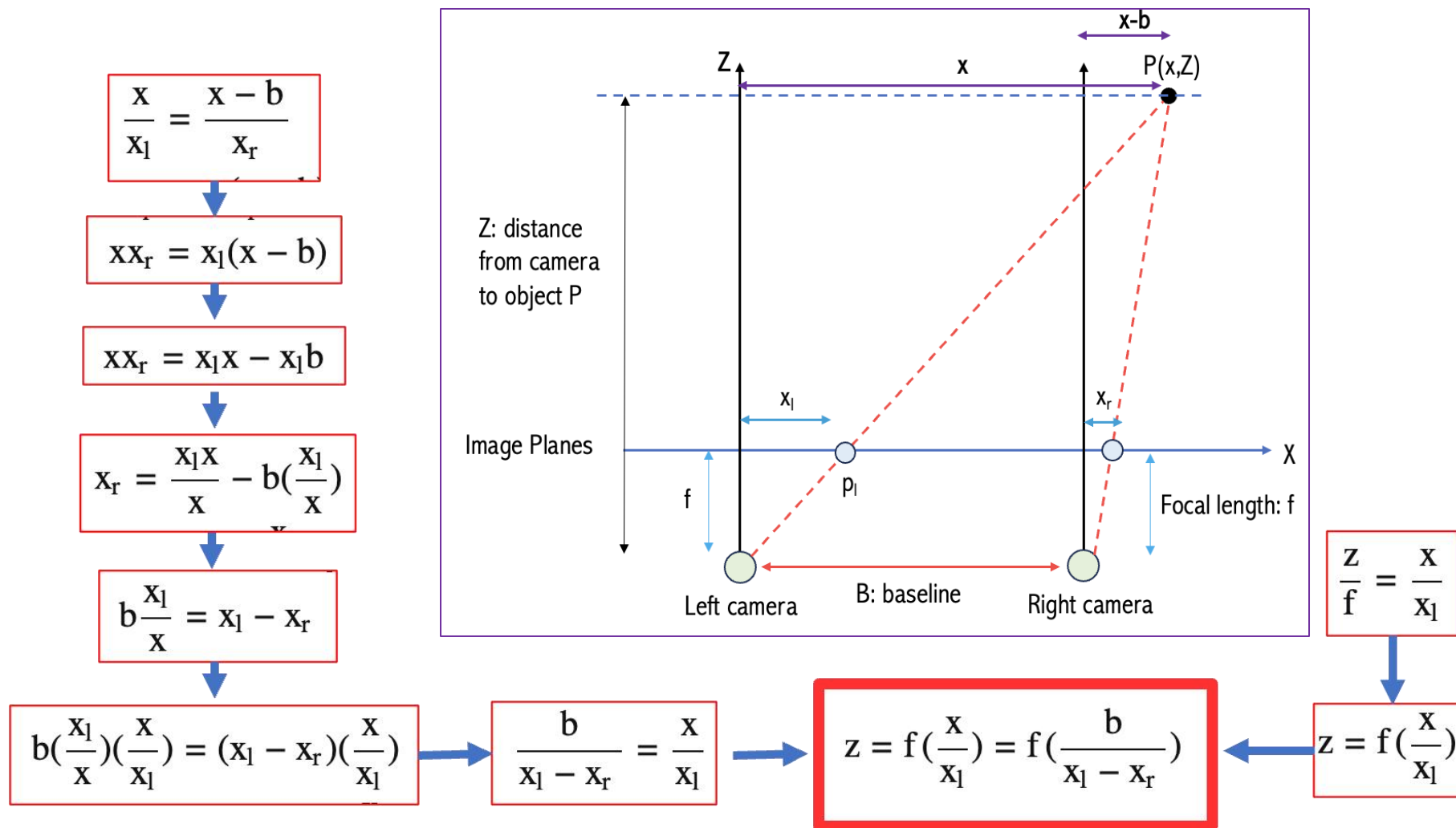
Stereo Matching: Background



$$\frac{Z}{f} = \frac{x}{x_l}$$

$$\frac{Z}{f} = \frac{x-b}{x_r}$$

Stereo Matching: Background



CASE STUDY



Object

Left image

255	254	253	252	251
15	16	17	255	255
18	18	17	255	255
15	17	18	255	255
255	255	255	255	255

Right image

255	254	253	252	251
255	15	18	18	255
255	16	17	20	255
255	14	14	19	255
255	255	255	255	255

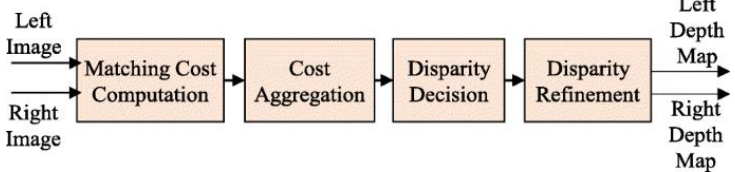
HOW?

1	1	1		
1	1	1		
1	1	1		

Left
Disparity

	1	1	1	
	1	1	1	
	1	1	1	

Right Disparity



CASE STUDY



Object

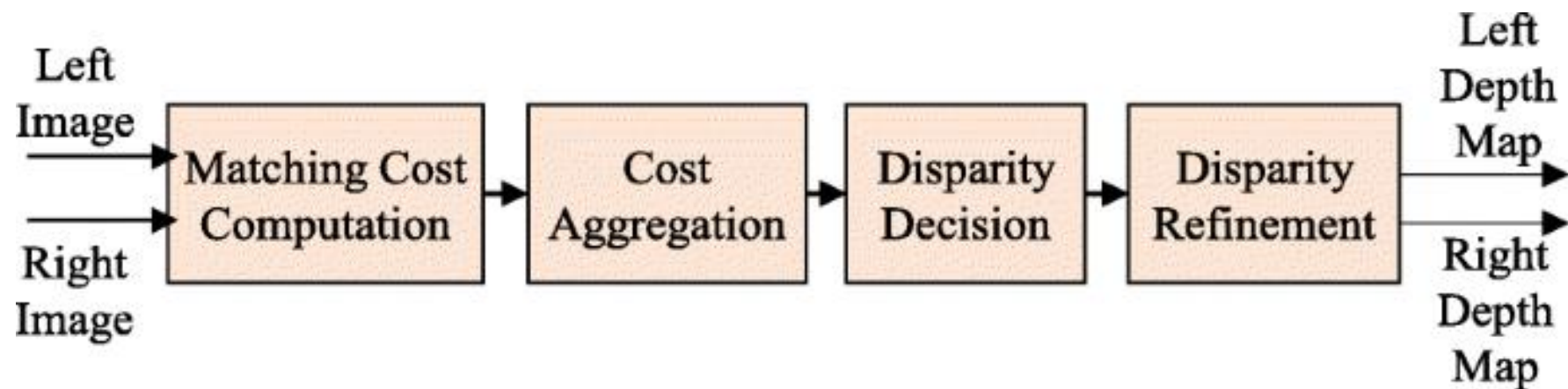
Left image

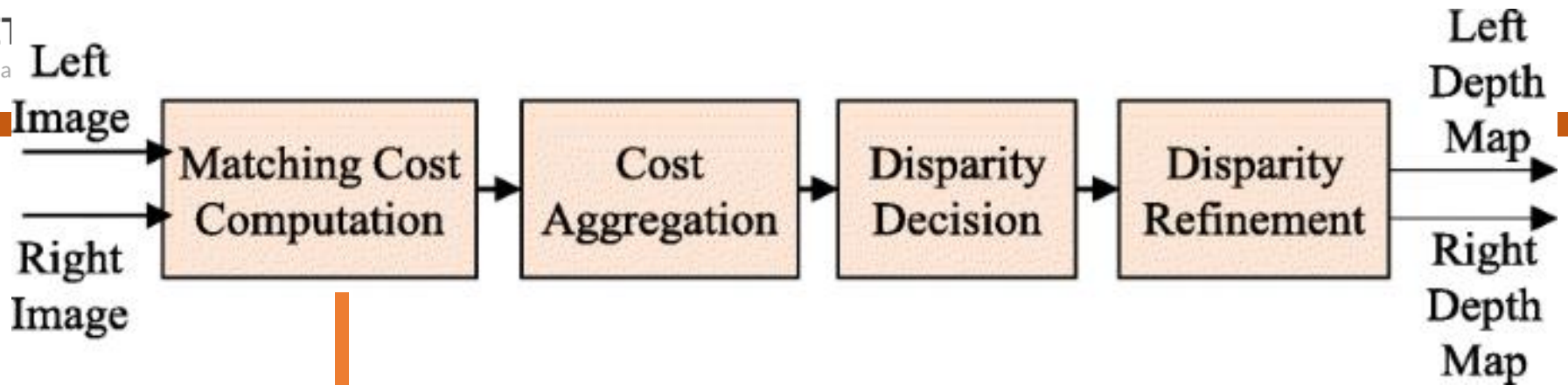
Right image

255	254	253	252	251
15	16	17	255	255
18	18	17	255	255
15	17	18	255	255
255	255	255	255	255

255	254	253	252	251
255	15	18	18	255
255	16	17	20	255
255	14	14	19	255
255	255	255	255	255

Choosing the direction of scientific research





Left image

255	254	253	252	251
15	16	17	255	255
18	18	17	255	255
15	17	18	255	255
255	255	255	255	255

Right image

255	254	253	252	251
255	15	18	18	255
255	16	17	20	255
255	14	14	19	255
255	255	255	255	255

Maximum Disparity/ Maximum Search Region: 3

Disparity	Cost
1	15-255
2	15-15
3	15-8

Cost
15-255
15-15
15-8



Z



Left image

255	254	253	252	251
15	16	17	255	255
18	18	17	255	255
15	17	18	255	255
255	255	255	255	255

Right image

255	254	253	252	251
255	15	18	18	255
255	16	17	20	255
255	14	14	19	255
255	255	255	255	255

Cost matrix size:

5x5x3

Maximum Disparity/
Maximum Search Region: 3

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Outline



➤ **How to read a research paper**

➤ **How to develop research questions**

➤ **Study and Implement Existing Methods**

➤ **How to develop a research plan**

➤ **Implement Your Proposed Method**

➤ **Performance Evaluation**

➤ **Enough Contributions for Submission**

Study and Implement Existing Methods

Define Your Research Question or Topic



Background Research



Study and Implement Existing Methods



Develop a Research Plan



Implement Your Proposed Methods



Evaluate Performance of Compared Methods



Stereo Matching

❖ Method 1: Implementation

$$d = \begin{bmatrix} d \\ 0 \end{bmatrix} \quad d \in D$$

$$C_1(p, d) = |L(p) - R(p - d)|$$

Finding d so that $C_1(p, d)$ is minimum.

$$d = \underset{d \in D}{\operatorname{argmin}}(C_1(p, d))$$

Then, d is the value for the pixel p in disparity map

```

16 # disparity map
17 depth = np.zeros((height, width), np.uint8)
18 scale = 255 / disparity_range
19
20 for y in range(height):
21     for x in range(width):
22
23         # find argmin_d
24         disparity = 0
25         cost_min = abs(left[y, x] - right[y, x])
26         for d in range(disparity_range):
27             if (x - d) < 0:
28                 cost = 255
29             else:
30                 cost = abs(left[y, x] - right[y, x - d])
31
32             # update cost_min
33             if cost < cost_min:
34                 cost_min = cost
35                 disparity = d
36
37         # set to the disparity map
38         depth[y, x] = disparity*scale
39
40 # save
41 cv2.imwrite('images/disparity_ad.png', depth)

```

Stereo Matching

❖ Method 1: Vectorization

$$\mathbf{d} = \begin{bmatrix} d \\ 0 \end{bmatrix} \quad d \in D$$

$$C_1(\mathbf{p}, \mathbf{d}) = |L(\mathbf{p}) - R(\mathbf{p} - \mathbf{d})|$$

Finding d so that $C_1(\mathbf{p}, d)$ is minimum.

$$d = \underset{d \in D}{\operatorname{argmin}}(C_1(\mathbf{p}, d))$$

Then, d is the value for the pixel $\mathbf{p}$ in disparity map

```

3 left = cv2.imread(left_img, 0)
4 right = cv2.imread(right_img, 0)
5
6 left = left.astype(np.float32)
7 right = right.astype(np.float32)
8
9 # compute cost space
10 cost_space = np.zeros((height, width, d_range))
11 for d in range(disparity_range):
12     left_d = left[:, d:width]
13     right_d = right[:, 0:width-d]
14     cost_d = np.abs(left_d-right_d)
15     cost_space[:, d:width, d] = cost_d
16
17 # compute disparity from cost space
18 scale = 255 / disparity_range
19 depth = np.argmin(cost_space, axis=2)
20 depth = depth*scale
21 depth = depth.astype(np.uint8)
22
23 # save
24 cv2.imwrite('images/disparity_ad.png', depth)

```


Stereo Matching

❖ Method 2

L is the left image

R is the right image

$L(\mathbf{p})$ is the (vector) value of $\mathbf{p}$

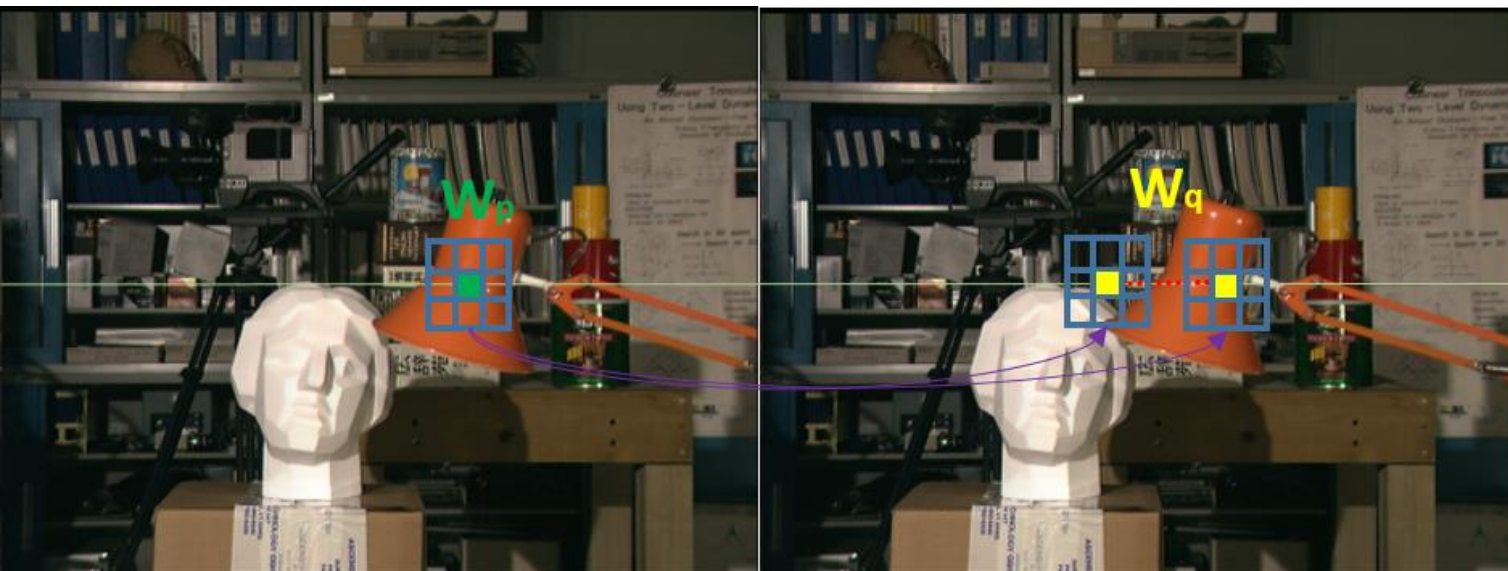
$$\mathbf{p} = \begin{bmatrix} x_p \\ y_p \end{bmatrix}$$

W_p is a window centered at $\mathbf{p}$

$$\mathbf{d} = \begin{bmatrix} d \\ 0 \end{bmatrix}$$

$$C_2(\mathbf{p}, \mathbf{d}) = \sum_{\mathbf{u} \in W_p} |L(\mathbf{u}) - R(\mathbf{u} - \mathbf{d})|$$

$$d \in D$$



Left Image

Right Image

Finding d so that $C_2(\mathbf{p}, \mathbf{d})$ is minimum.

$$d = \underset{d \in D}{\operatorname{argmin}} (C_2(\mathbf{p}, \mathbf{d}))$$

Then, d is the value for the pixel $\mathbf{p}$ in disparity map

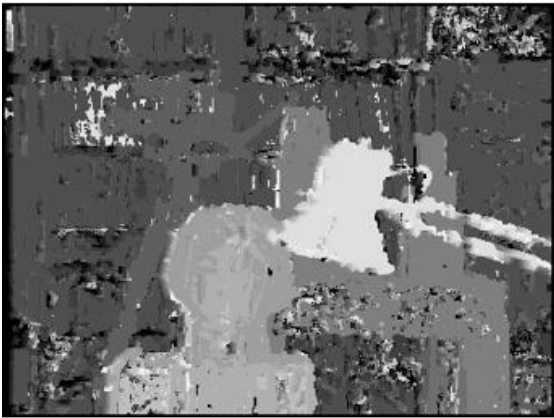
Stereo Matching



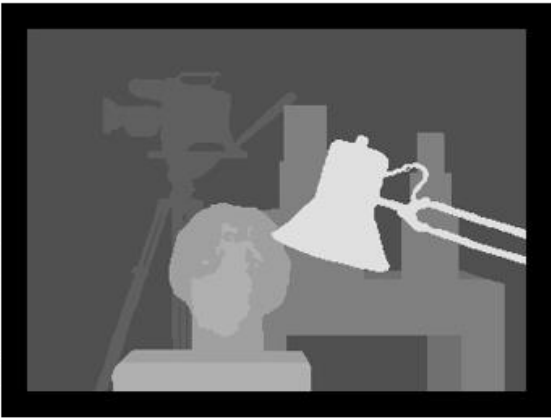
Left Image



Right Image



Disparity Map



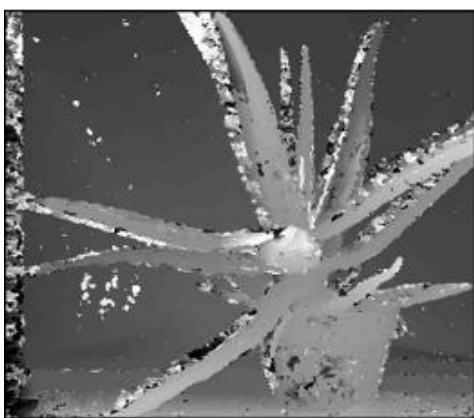
Ground Truth



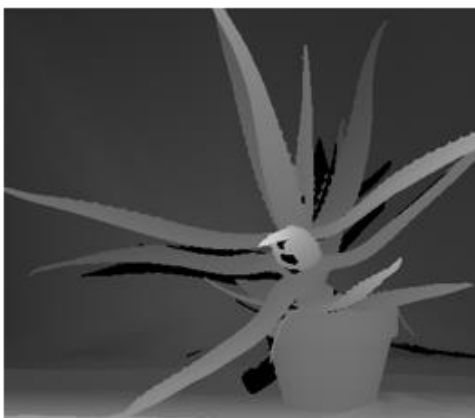
Left Image



Right Image



Disparity Map



Ground Truth

Outline

- **How to read a research paper**
- **How to develop research questions**
- **Study and Implement Existing Methods**
- **How to develop a research plan**
- **Implement Your Proposed Method**
- **Performance Evaluation**
- **Enough Contributions for Submission**



Develop Research Plan



- 1. Research Title:**
Provide a clear and concise title that reflects the main focus of your research
- 2. Introduction**
Briefly introduce your research topic, explaining its significance and relevance.
- 3. Background and Literature Review**
Summarize the existing knowledge and research related to your topic. Highlight key studies, theories, or gaps in the literature that your research aims to address.
- 4. Research Objectives**
What do you aim to achieve or discover through your study?
- 5. Research Questions**
Formulate the research questions or hypotheses that will guide your investigation
- 6. Methodology**
Detail the research design, methods, and procedures you plan to employ
- 7. Timeline**
Provide a timeline for your research activities. Include milestones, deadlines, and key events.
- 8. Resources**
Outline the resources you'll need for your research, such as access to specific databases, equipment, or funding.
- 9. Expected Result**
Discuss the anticipated outcomes of your research.
- 10. Significant and Contribution**
Explain the potential significance of your research.

Develop Research Plan



1. Research Title:

Robust Stereo Matching Various Weather Conditions

2. Introduction

Stereo matching is a fundamental computer vision task that involves the computation of dense disparity maps from stereo image pairs. The goal is to establish correspondences between pixels in the left and right images, providing depth information crucial for 3D perception. However, the accuracy and reliability of stereo matching algorithms are often challenged by various real-world factors such as occlusions, varying lighting conditions, and textureless regions.

In response to these challenges, the concept of robust stereo matching has emerged as a critical area of research within computer vision. Robust stereo matching aims to enhance the

Develop Research Plan



1. Research Title: (done)

2. Introduction

...

Robust stereo matching techniques incorporate advanced algorithms and methodologies to address common issues encountered in stereo vision. These may include the development of adaptive matching cost functions, the integration of machine learning approaches for robust feature extraction, and the incorporation of strategies to handle occlusions and textureless regions effectively.

This field plays a pivotal role in applications such as autonomous vehicles, robotics, and augmented reality, where precise depth perception is essential for making informed decisions. As researchers delve into the complexities of real-world stereo vision challenges, the development of robust stereo matching algorithms becomes increasingly crucial to unlocking the full potential of stereo imaging systems in diverse and dynamic environments.

Develop Research Plan



1. Research Title (done)
2. Introduction (done)
3. Background and Literature Review

3.a Introduction

Stereo matching, a vital component of computer vision, aims to extract depth information from stereo image pairs. This literature review explores the historical evolution, foundational concepts, and recent advancements in stereo matching algorithms.

3.b Historical of Stereo Matching

Early stereo matching methods, such as block matching and dynamic programming, laid the groundwork for depth perception. Despite their simplicity, these approaches struggled with challenges like occlusions and varying lighting conditions.

3. Foundational Concepts in Stereo Matching

Key concepts, including epipolar geometry and the stereo correspondence problem, form the basis of stereo matching. Understanding these

Develop Research Plan



1. **Research Title (done)**
2. **Introduction (done)**
3. **Background and Literature Review**

3.d Traditional Stereo Matching Techniques

Classical methods like block-based matching and graph cuts provided initial solutions to stereo correspondence. However, they faced limitations in handling textureless regions and were sensitive to noise.

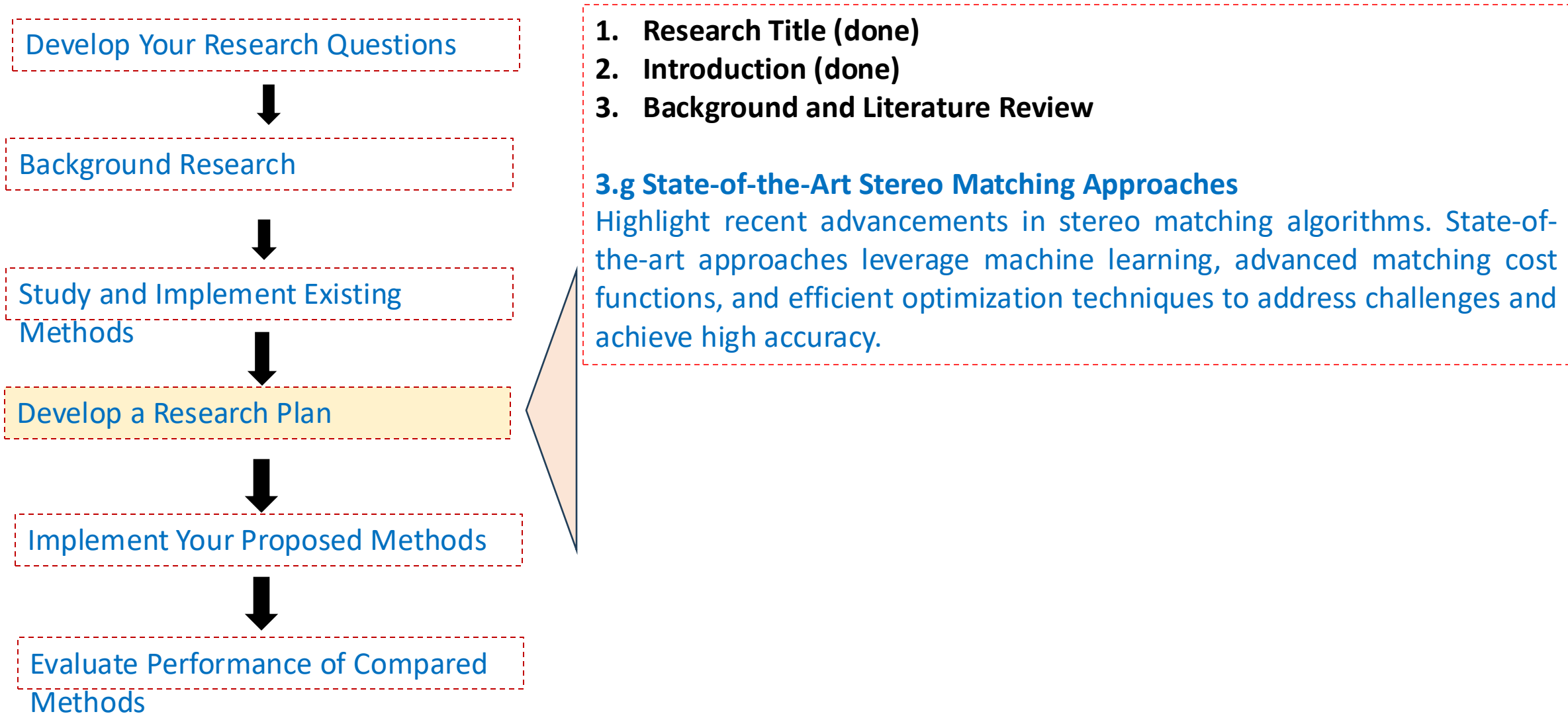
3.e Historical of Stereo Matching

Early stereo matching methods, such as block matching and dynamic programming, laid the groundwork for depth perception. Despite their simplicity, these approaches struggled with challenges like occlusions and varying lighting conditions.

3.f Challenges in Traditional Stereo Matching

Discuss challenges faced by traditional stereo matching techniques, such as sensitivity to noise, computational inefficiency, and difficulties in handling real-world scenarios with dynamic lighting conditions.

Develop Research Plan



Develop Research Plan



1. **Research Title (done)**
2. **Introduction (done)**
3. **Background and Literature Review (done)**
4. **Research Objectives**

Investigate and address challenges such as occlusions, varying lighting conditions, and textureless regions. Research could focus on developing algorithms that are robust in real-world scenarios, ensuring reliable depth estimation even in adverse conditions.

Explore the integration of machine learning, including deep learning techniques, to enhance feature extraction and matching cost computation. This objective aims to leverage the power of neural networks to improve the overall performance of stereo matching algorithms.

Develop real-time stereo matching algorithms for applications that require fast and efficient processing, such as robotics or autonomous vehicles. This objective involves optimizing algorithms for speed without

Develop Research Plan



1. **Research Title**
2. **Introduction**
3. **Background and Literature Review**
4. **Research Objectives**
5. **Research Questions**

RQ1: How to improve the performance of existing stereo matching methods under various lighting conditions and complex material properties?

RQ2: How to obtain a real-time processing for current state-of-the-art stereo matching-based deep learning methods?

RQ3: How to design an automatically learning the network architecture, its activation functions, and its parameters from data ?

Develop Research Plan



1. **Research Title:**
2. **Introduction**
3. **Background and Literature Review**
4. **Research Objectives**
5. **Research Questions**
6. **Methodology**

a. Data Preprocessing:

Noise Reduction: Apply preprocessing techniques to reduce noise in the stereo images. This may include Gaussian blurring, median filtering, or other denoising methods to improve the quality of input data.

Color Correction: Normalize lighting conditions and correct color disparities between the stereo images to ensure consistency in feature extraction.

b. Feature Extraction:

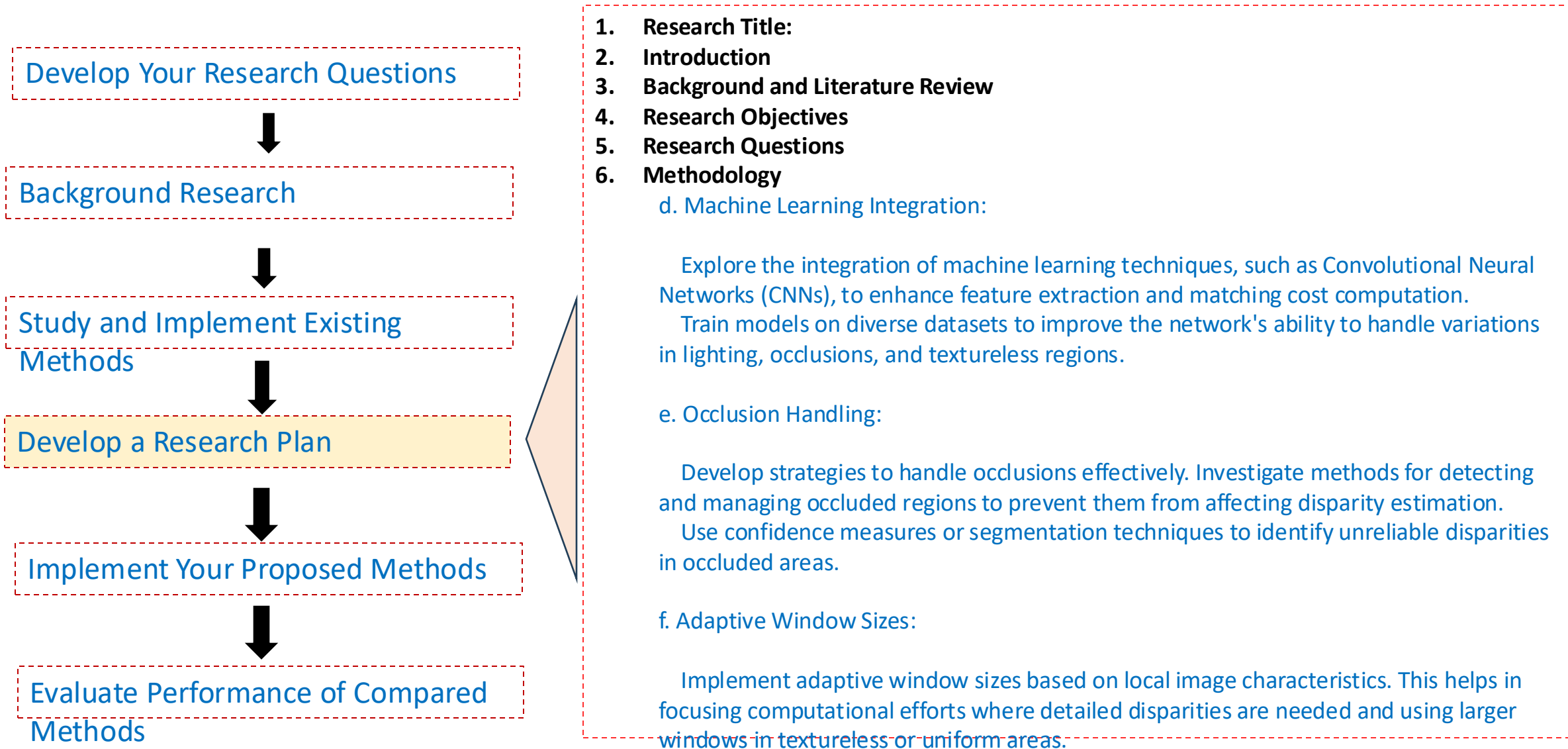
Use advanced feature extraction techniques to capture meaningful information from the stereo images.

Explore local features (e.g., SIFT, SURF) and global features (e.g., color histograms) that are robust to variations in lighting and texture.

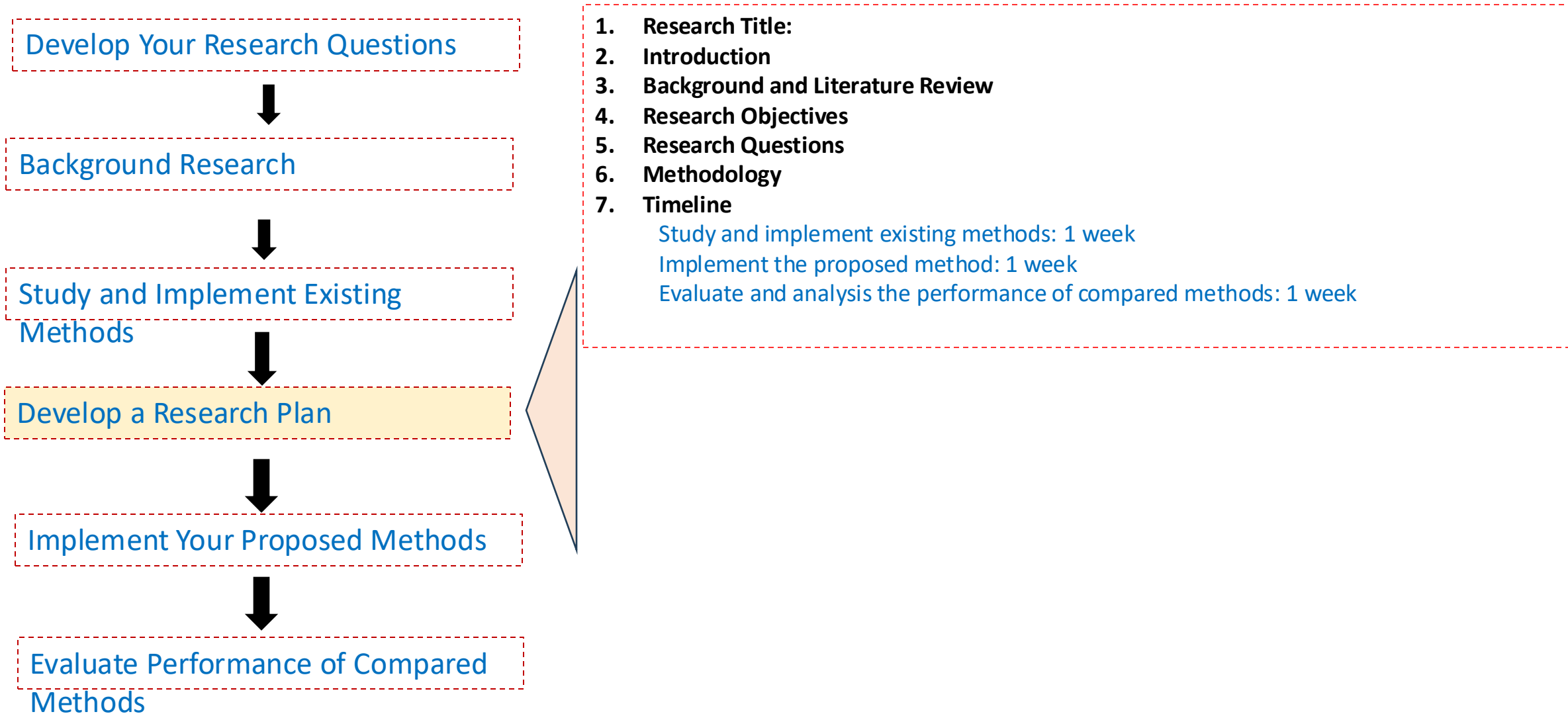
c. Matching Cost Computation:

Employ adaptive matching cost functions that dynamically adjust to local image characteristics.

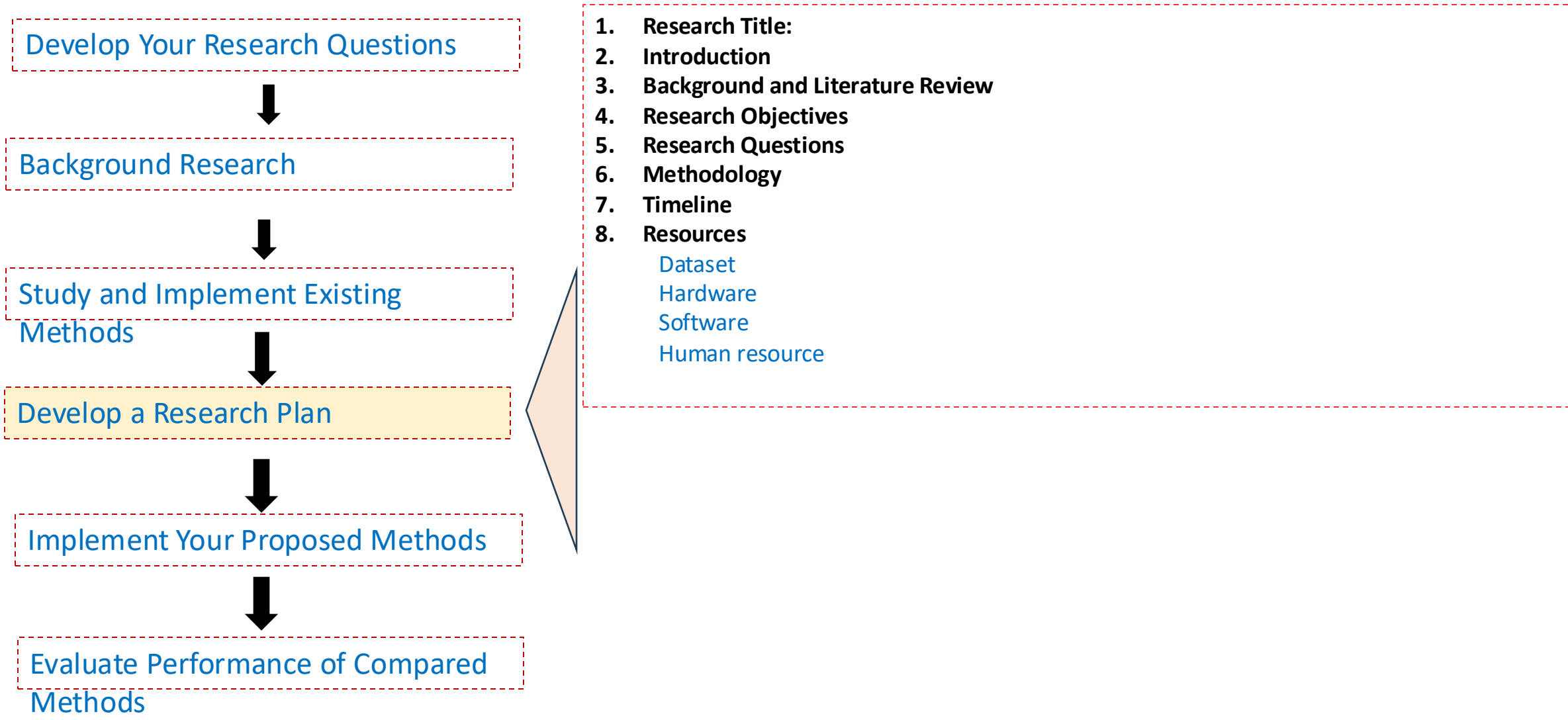
Develop Research Plan



Develop Research Plan



Develop Research Plan



Develop Research Plan



1. **Research Title:**
2. **Introduction**
3. **Background and Literature Review**
4. **Research Objectives**
5. **Research Questions**
6. **Methodology**
7. **Timeline**
8. **Resources**
9. **Expected Result**
 - Improve the performance of existing methods under difficult conditions by 1%.
 - Reduce the processing time of the-state-of-the-art method to 30ms/frame

Develop Research Plan



1. **Research Title:**
2. **Introduction**
3. **Background and Literature Review**
4. **Research Objectives**
5. **Research Questions**
6. **Methodology**
7. **Timeline**
8. **Resources**
9. **Expected Result**
10. **Significant and Contribution**

Significance: The proposed stereo matching methodology addresses the limitations of existing techniques by specifically targeting challenging conditions, including occlusions, varying lighting conditions, and textureless regions.

Contribution: The methodology introduces novel strategies, such as adaptive matching cost functions and machine learning integration, to enhance the accuracy and robustness of disparity estimation under difficult scenarios.

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Implement Your Proposed Method

RQ1: How to improve the performance of existing stereo matching methods under various lighting conditions and complex material properties?

RQ2: How to obtain a real-time processing for current state-of-the-art stereo matching-based deep learning methods?

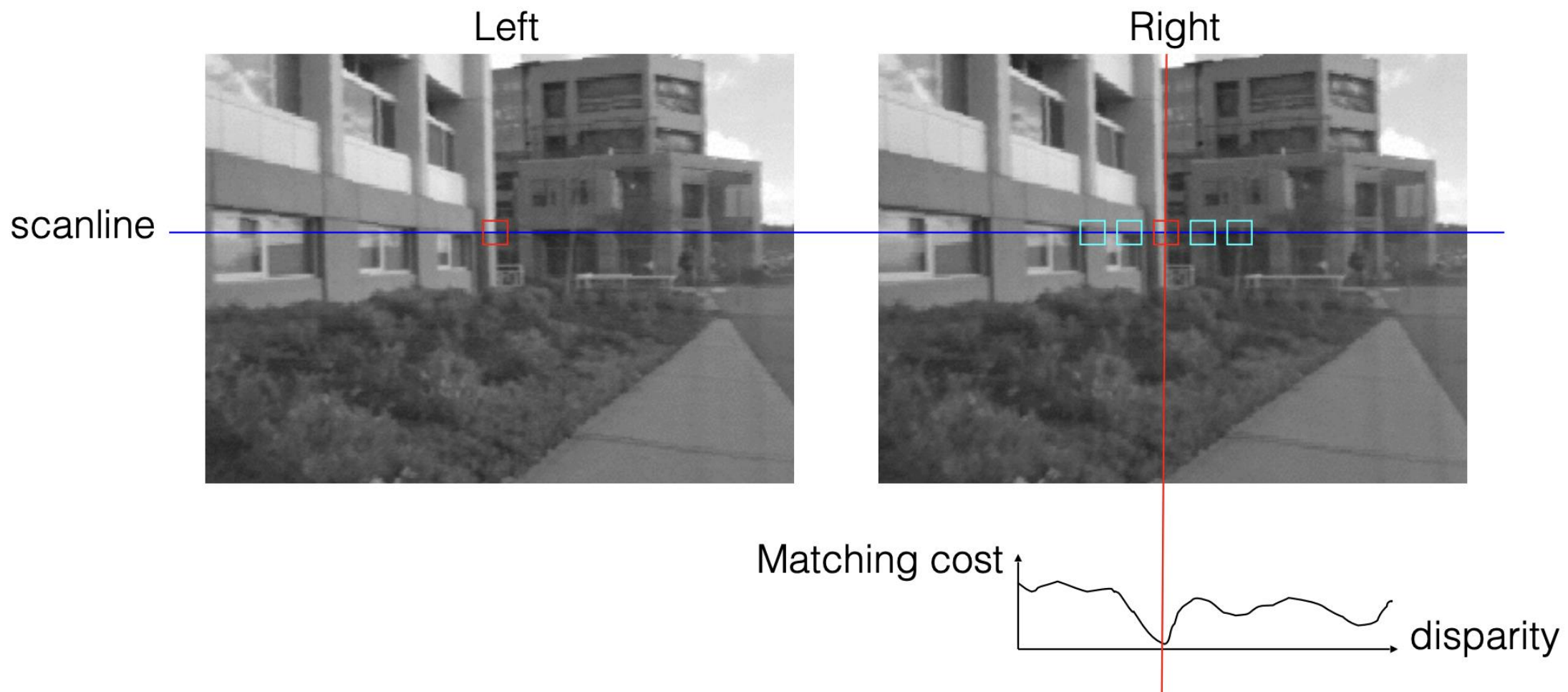
RQ3: How to design an automatically learning the network architecture, its activation functions, and its parameters from data ?

Please select the simplest question to solve



In this module

Stereo Matching: Background



Stereo Matching: Background

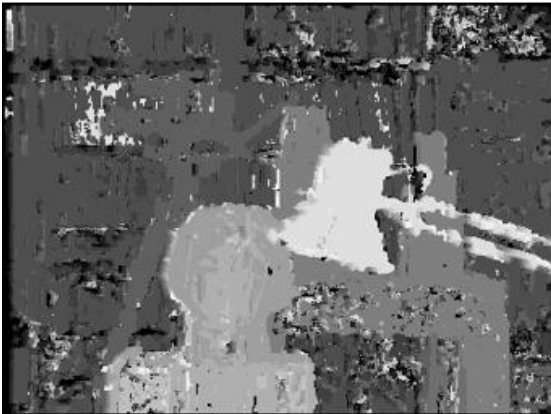
❖ Normal



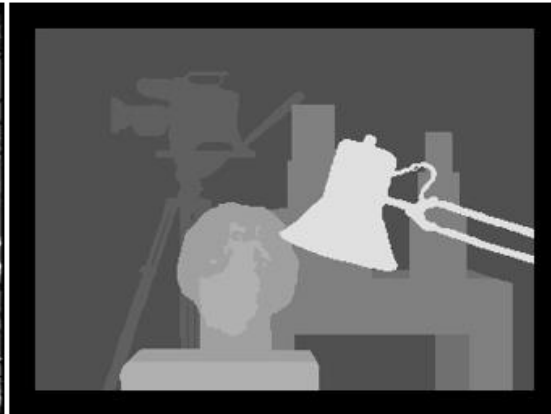
Left Image



Right Image



Disparity Map



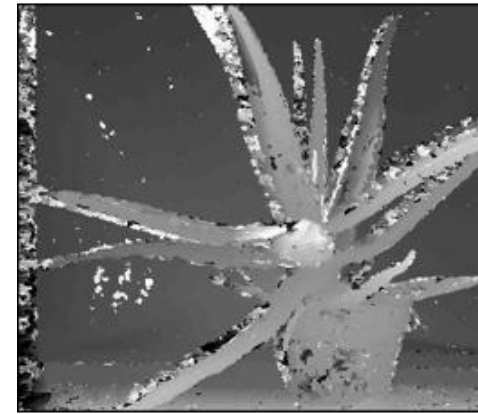
Ground Truth



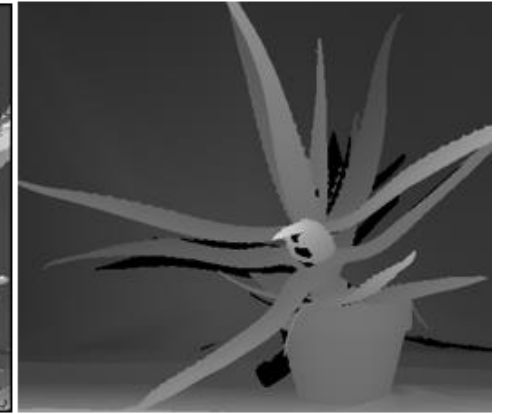
Left Image



Right Image



Disparity Map



Ground Truth

Stereo Matching: Background

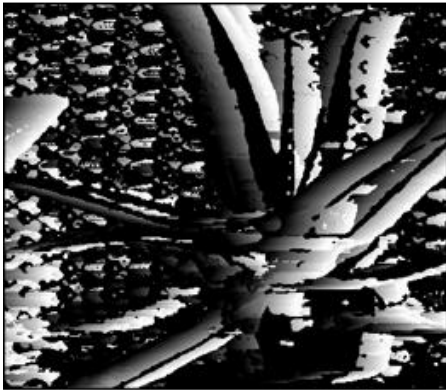
❖ Challenges



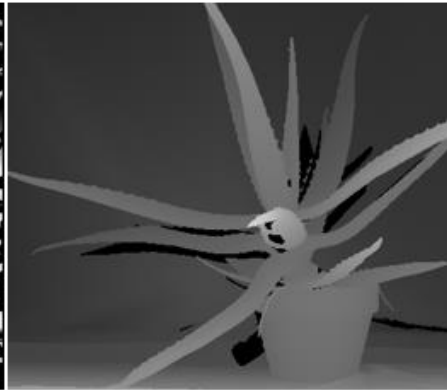
Left Image



Right Image



Disparity Map



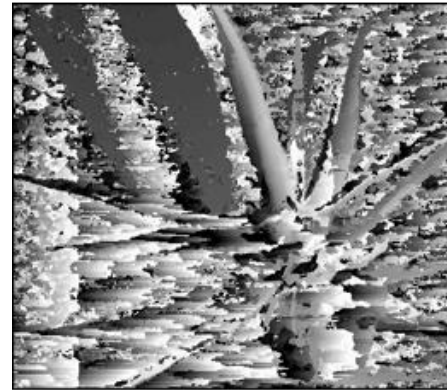
Ground Truth



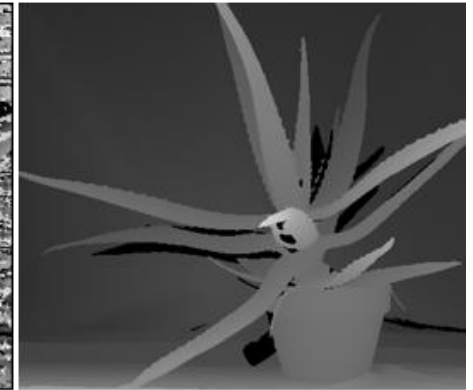
Left Image



Right Image



Disparity Map



Ground Truth



Stereo Matching: Background

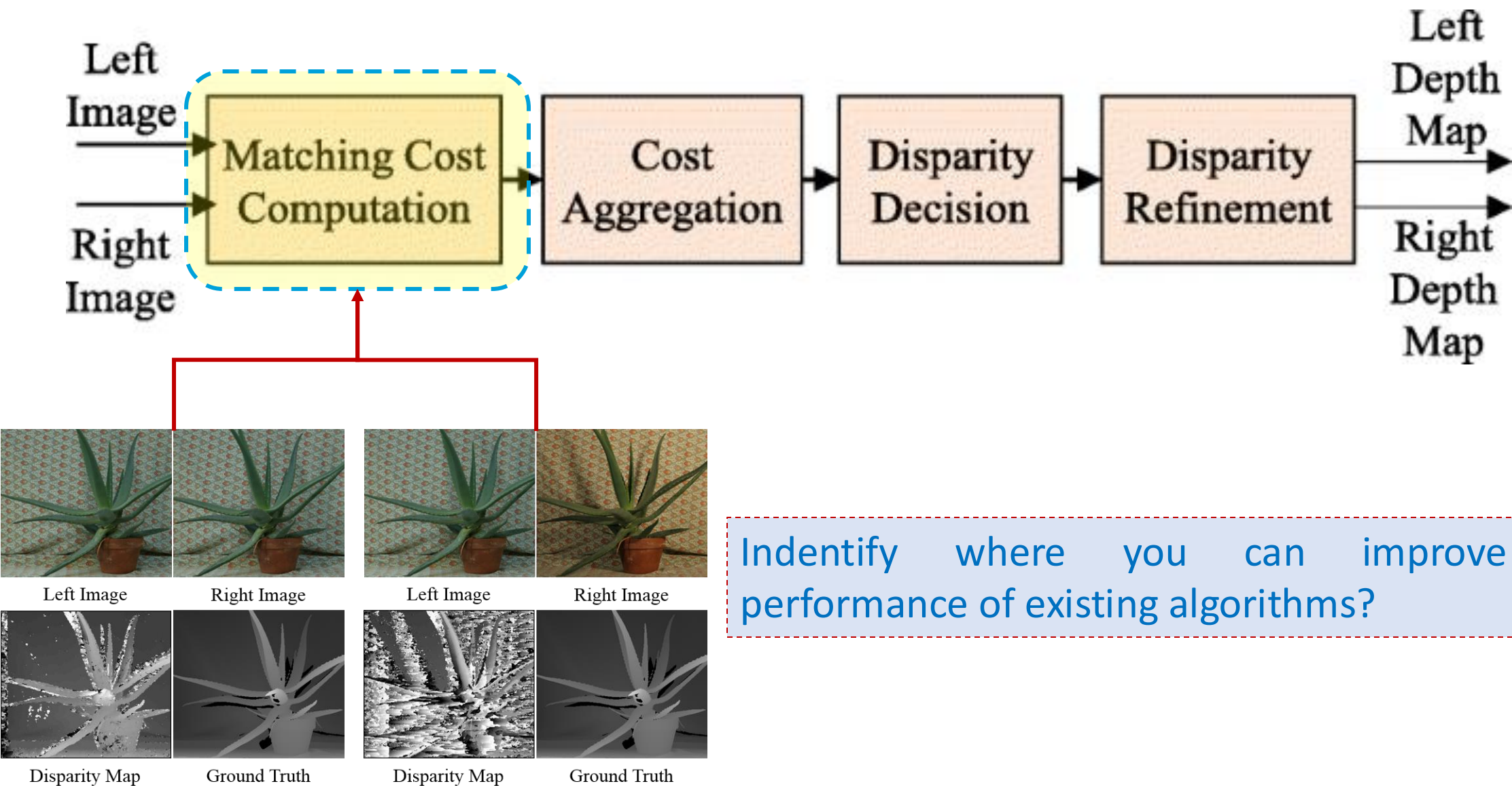
❖ Challenges



Photometric
Variations.



Stereo Matching Workflow



Stereo Matching Cost

Parametric Matching Costs

$$C_{AD}(\mathbf{p}, \mathbf{d}) = |I_L(\mathbf{p}) - I_R(\mathbf{p} - \mathbf{d})|, \quad (1)$$

$$C_{SAD}(\mathbf{p}, \mathbf{d}) = \sum_{\mathbf{q} \in N_{\mathbf{p}}} |I_L(\mathbf{q}) - I_R(\mathbf{q} - \mathbf{d})|. \quad (2)$$

Sampling-insensitive
absolute difference of
Birchfield and Tomasi

$$C_{BT}(\mathbf{p}, \mathbf{d}) = \min(A, B),$$

$$A = \max(0, I_L(\mathbf{p}) - I_R^{max}(\mathbf{p} - \mathbf{d}), I_R^{min}(\mathbf{p} - \mathbf{d}) - I_L(\mathbf{p})),$$

$$B = \max(0, I_R(\mathbf{p} - \mathbf{d}) - I_L^{max}(\mathbf{p}), I_L^{min}(\mathbf{p}) - I_R(\mathbf{p} - \mathbf{d})),$$

$$I^{min}(\mathbf{p}) = \min(I^-(\mathbf{p}), I(\mathbf{p}), I^+(\mathbf{p}))$$

$$I^{max}(\mathbf{p}) = \max(I^-(\mathbf{p}), I(\mathbf{p}), I^+(\mathbf{p}))$$

$$I^-(\mathbf{p}) = (I(\mathbf{p} - [1 \ 0]^T) + I(\mathbf{p}))/2,$$

$$I^+(\mathbf{p}) = (I(\mathbf{p} + [1 \ 0]^T) + I(\mathbf{p}))/2.$$

Cosine similarity

Cosine similarity (cs) được dùng để đo mức độ giống nhau/tương đồng giữa hai vector

Gọi $\vec{x}$ và $\vec{y}$ là hai vector, cs được tính như sau

$$cs(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_1^n x_i y_i}{\sqrt{\sum_1^n x_i^2} \sqrt{\sum_1^n y_i^2}}$$

Tính chất 1: $cs(\vec{x}, \vec{y}) = cs(a\vec{x}, b\vec{y})$

$$\begin{aligned} cs(a\vec{x}, b\vec{y}) &= \frac{a\vec{x} \cdot b\vec{y}}{\|a\vec{x}\| \|b\vec{y}\|} = \frac{\sum_1^n a x_i b y_i}{\sqrt{\sum_1^n a^2 x_i^2} \sqrt{\sum_1^n b^2 y_i^2}} \\ &= \frac{ab \sum_1^n x_i y_i}{\sqrt{a^2 \sum_1^n x_i^2} \sqrt{b^2 \sum_1^n y_i^2}} \\ &= \frac{\sum_1^n x_i y_i}{\sqrt{\sum_1^n x_i^2} \sqrt{\sum_1^n y_i^2}} = cs(\vec{x}, \vec{y}) \end{aligned}$$

Tính chất 2: $cs(\vec{x}, \vec{y}) \neq cs(\vec{x} + c, \vec{y} + d)$

Ví dụ: $\vec{x} = [4, 2, 1, 2]^T$

$$\vec{y} = [1, 2, 2, 0]^T$$

$$\vec{u} = 2\vec{x} = [8, 4, 2, 4]^T$$

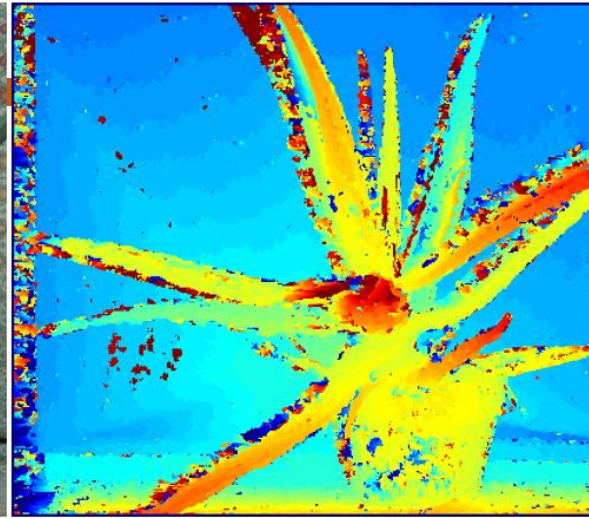
$$\vec{v} = 3\vec{y} = [3, 6, 6, 0]^T$$

$$\begin{aligned} cs(\vec{x}, \vec{y}) &= \frac{4*1 + 2*2 + 1*2 + 2*0}{\sqrt{4^2 + 2^2 + 1^2 + 2^2} \sqrt{1^2 + 2^2 + 2^2 + 0}} \\ &= \frac{10}{\sqrt{25} \sqrt{9}} = \frac{10}{15} = 0.67 \end{aligned}$$

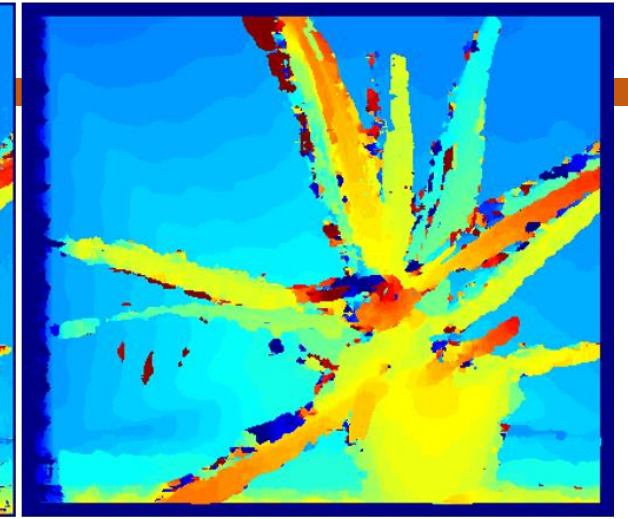
$$\begin{aligned} cs(\vec{u}, \vec{v}) &= \frac{8*3 + 4*6 + 2*6 + 4*0}{\sqrt{8^2 + 4^2 + 2^2 + 4^2} \sqrt{3^2 + 6^2 + 6^2 + 0}} \\ &= \frac{60}{\sqrt{100} \sqrt{81}} = \frac{60}{90} = 0.67 \\ &= cs(\vec{x}, \vec{y}) \end{aligned}$$



Ảnh stereo có cùng độ sáng



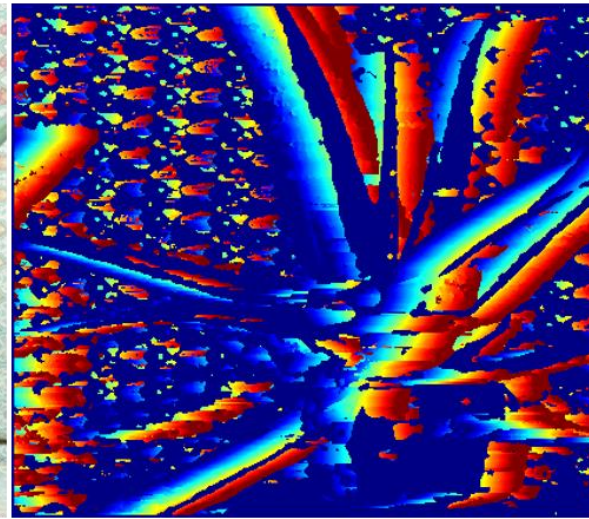
Absolute difference



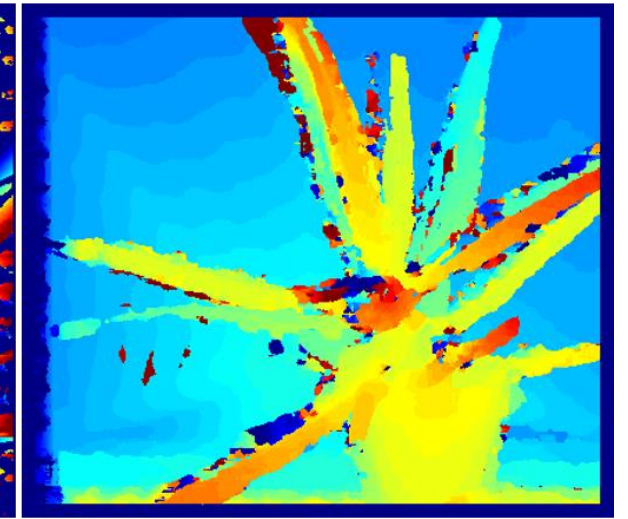
Cosine Similarity



Ảnh stereo khác độ sáng



Absolute difference



Cosine Similarity

Cosine similarity hoạt động ổn định khi ảnh stereo thay đổi độ sáng

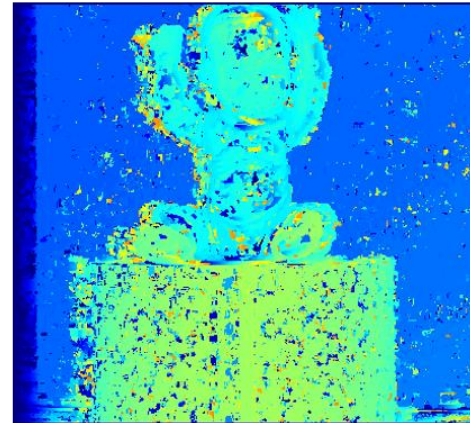
Correlation for Stereo Matching Cost



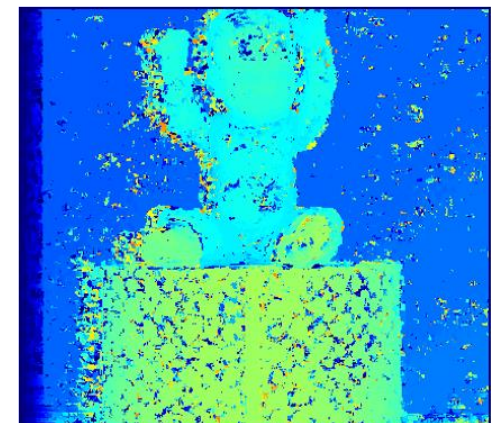
Left image



Right image



Cosine Result/ RMS = 5.6



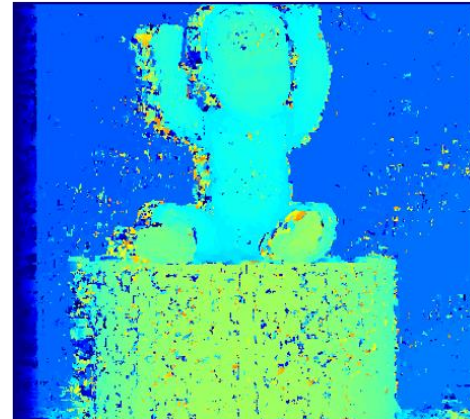
Correlation Result / RMS = 5.38



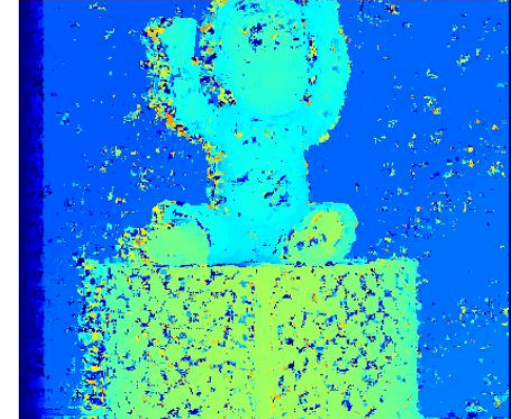
Left image



Right image



Cosine Result / RMS = 4.98



Correlation Result / RMS = 5.36

Outline

- **How to read a research paper**
- **How to develop research questions**
- **Study and Implement Existing Methods**
- **How to develop a research plan**
- **Implement Your Proposed Method**
- **Performance Evaluation**
- **Enough Contributions for Submission**



Datasets





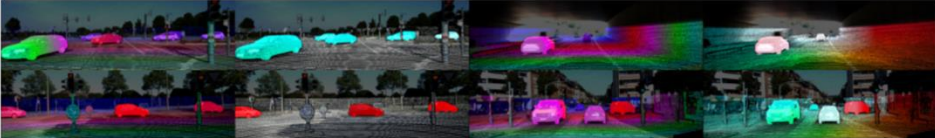
The KITTI Vision Benchmark Suite

A project of Karlsruhe Institute of Technology and Toyota Technological Institute at Chicago

home setup **stereo** flow sceneflow depth odometry object tracking road semantics raw data submit results

A. Geiger | P. Lenz | C. Stiller | R. Urtasun Log in

Stereo Evaluation 2015



The **stereo 2015** / **flow 2015** / **scene flow 2015** benchmark consists of 200 training scenes and 200 test scenes (4 color images per scene, saved in loss less png format). Compared to the stereo 2012 and flow 2012 benchmarks, it comprises dynamic scenes for which the ground truth has been established in a semi-automatic process. Our evaluation server computes the percentage of bad pixels averaged over all ground truth pixels of all 200 test images. For this benchmark, we consider a pixel to be correctly estimated if the disparity or flow end-point error is $< 3\text{px}$ or $< 5\%$ (for scene flow this criterion needs to be fulfilled for both disparity maps and the flow map). We require that all methods use the same parameter set for all test pairs. Our development kit provides details about the data format as well as MATLAB / C++ utility functions for reading and writing disparity maps and flow fields. More details can be found in [Object Scene Flow for Autonomous Vehicles \(CVPR 2015\)](#).

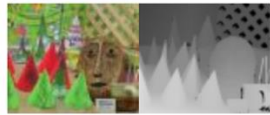
- Download stereo 2015/flow 2015/scene flow 2015 data set (2 GB)
- Download calibration files (1 MB)
- Download multi-view extension (20 frames per scene) (14 GB)
- Download development kit (3 MB)

Stereo
Evaluation
Datasets
Code
Submit

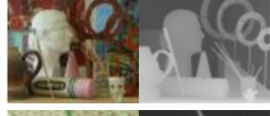
Middlebury Stereo Datasets



[2001 datasets](#) - 6 datasets of piecewise planar scenes [1]
(Sawtooth, Venus, Bull, Poster, Barn1, Barn2)



[2003 datasets](#) - 2 datasets with ground truth obtained using structured light [2]
(Cones, Teddy)



[2005 datasets](#) - 9 datasets obtained using the technique of [2], published in [3, 4]
(Art, Books, Dolls, Laundry, Moebius, Reindeer, Computer, Drumsticks, Dwarves)



[2006 datasets](#) - 21 datasets obtained using the technique of [2], published in [3, 4]
(Aloe, Baby1-3, Bowling1-2, Cloth1-4, Flowerpots, Lampshade1-2, Midd1-2, Monopoly, Plastic, Rocks1-2, Wood1-2)



[2014 datasets](#) - 33 datasets obtained using the technique of [5]



[2021 mobile datasets](#) - 24 datasets obtained with a mobile device on a robot arm, using the technique of [5]

Evaluation Metrics

	Method	Setting	Code	D1-bg	D1-fg	<u>D1-all</u>	Density	Runtime	Environment
1	<u>MoCha-Stereo</u>			1.36 %	2.43 %	1.53 %	100.00 %	0.27 s	NVIDIA Tesla A6000 (PyTorch)
2	<u>DiffuVolume</u>			1.35 %	2.51 %	1.54 %	100.00 %	0.36 s	GPU @ 2.5 Ghz (Python)
3	<u>GANet+ADL</u>			1.38 %	2.38 %	1.55 %	100.00 %	0.67s	NVIDIA RTX 3090 (PyTorch)
4	<u>Selective-IGEV</u>			1.33 %	2.61 %	1.55 %	100.00 %	0.24 s	1 core @ 2.5 Ghz (Python)
5	<u>MC-Stereo</u>			1.36 %	2.51 %	1.55 %	100.00 %	0.40 s	1 core @ 2.5 Ghz (Python)
6	<u>IGEV-ICGNet</u>			1.38 %	2.55 %	1.57 %	100.00 %	0.18 s	NVIDIA Tesla A5000 (Pytorch)
7	<u>yjlig</u>			1.37 %	2.62 %	1.58 %	100.00 %	0.35 s	1 core @ 2.5 Ghz (C/C++)
8	<u>MDA</u>			1.37 %	2.64 %	1.58 %	100.00 %	0.32 s	1 core @ 2.5 Ghz (Python)
9	<u>UGNet</u>			1.34 %	2.77 %	1.58 %	100.00 %	0.2 s	GPU @ 3.0 Ghz (Python)
10	<u>Any-IGEV</u>			1.43 %	2.35 %	1.58 %	100.00 %	0.32 s	GPU @ 2.5 Ghz (Python)
11	<u>OpenStereo-IGEV</u>		<u>code</u>	1.44 %	2.31 %	1.59 %	100.00 %	0.18 s	NVIDIA-3090



Evaluation Metrics

Percentage of Bad Pixels: Measures the percentage of pixels with disparity errors exceeding a specified threshold. Common thresholds include 1, 2, or 3 pixels.

$$BP = \frac{\text{Number of Bad Pixels}}{\text{Total Number of Pixels}} \times 100\%$$

Root Mean Squared Error (RMSE): Calculates the square root of the average squared disparity errors across all pixels. It provides a measure of the overall error.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (D_i - \hat{D}_i)^2}$$

Mean Absolute Error (MAE): Computes the average absolute disparity errors. It is less sensitive to outliers than RMSE.

$$MAE = \frac{1}{N} \sum_{i=1}^N |D_i - \hat{D}_i|$$

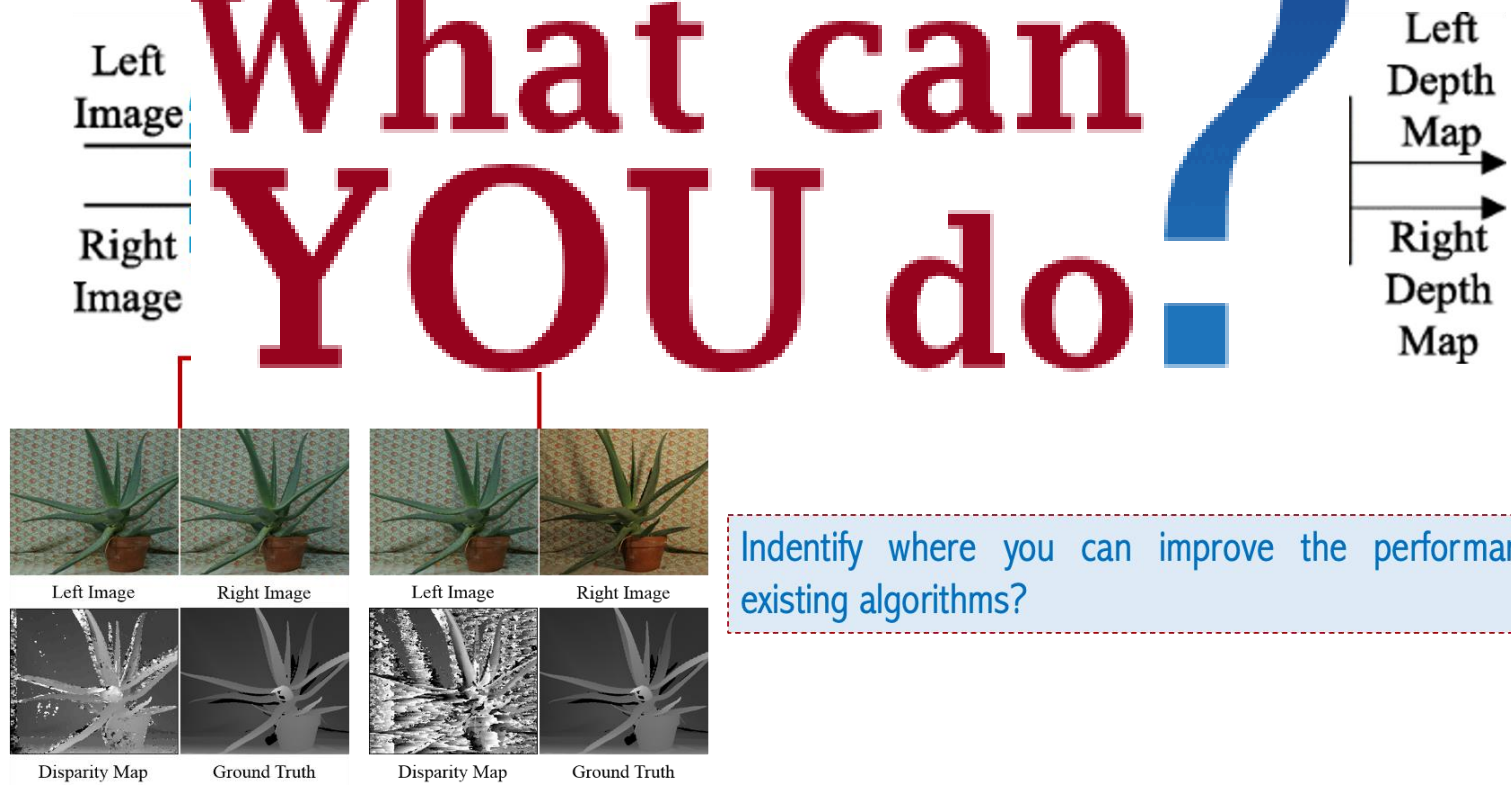
The "d1-bg" error is specifically defined as the percentage of disparity errors (in pixels) where the absolute disparity error is less than 1 pixel and the error occurs in regions classified as foreground or background

$$d1\text{-bg} = \frac{\text{Number of pixels with disparity error} < 1 \text{ pixel and in fg/bg}}{\text{Total number of pixels in fg/bg}} \times 100\%$$

Thorough Analysis

- Examine your research method and results critically. Identify the specific aspects needed. Understand the limitations are facing.

What can
YOU do?



Identify where you can improve the performance of existing algorithms?

Fail to Improve Performance of Existing Methods

Thorough Analysis

- Examine your research method and results critically. Identify the specific aspects needed. Understand the limitations are facing.

Literature Review

- Revisit the literature to ensure you are aware of the field. Identify if there are that could be applied to your

Thorough Analysis

- Seek feedback from experts, colleagues, or mentors. Discuss for their insights. They may offer alternative approaches.

Work with Peers

Researchers working on similar perspectives. Sharing ideas challenges can lead to

Consider a Multidisciplinary Approach:

- Explore whether incorporating knowledge from other disciplines could benefit your research. Sometimes, a multidisciplinary approach can provide novel solutions.

What can YOU do?
Innovative Solutions.

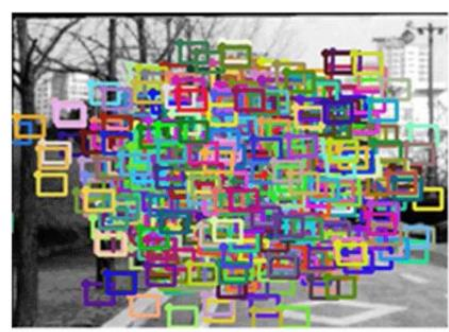
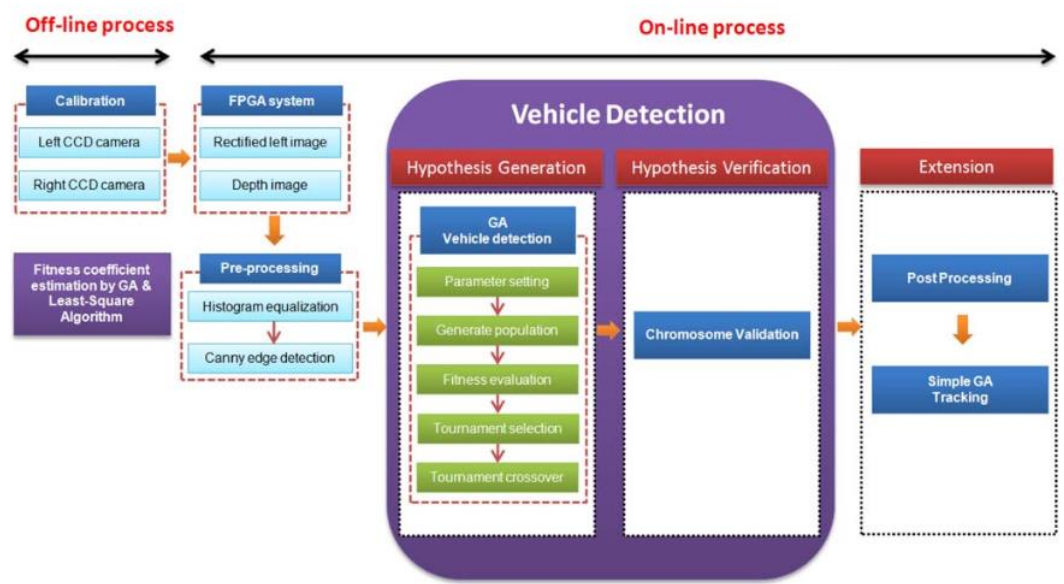
Fail to Improve Performance of Existing Methods

Consider a Multidisciplinary Approach:

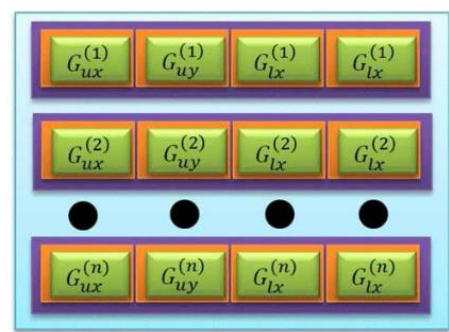
- Explore whether incorporating knowledge from other disciplines could benefit your research. Sometimes, a multidisciplinary approach can provide novel solutions.



Fig. 3. Proposed system with two charge-coupled device cameras, a chess-board, an FPGA board, and a PC for stereo camera calibration.



(a)



(b)

Population Chromosome Building Block Gene

Fig. 4. (a) Initial population based on a Gaussian process. (b) Chromosome structure.

V. D. Nguyen, T. T. Nguyen, D. D. Nguyen, S. J. Lee and J. W. Jeon, "A Fast Evolutionary Algorithm for Real-Time Vehicle Detection," in IEEE Transactions on Vehicular Technology, vol. 62, no. 6, pp. 2453-2468, July 2013, doi: 10.1109/TVT.2013.2242910.

Fail to Improve Performance of Existing Methods

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e with Peers

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- Explore whether incorporating knowledge from other disciplines could benefit your research. Sometimes, a multidisciplinary approach can provide novel solutions.

Experiment with Variations

- Try variations of your existing method to identify potential improvements. Experiment with different parameters, data preprocessing techniques, or algorithms to see if any adjustments lead to better results.

**What can
YOU do?**



Fail to Improve Performance of Existing Methods

Validation and Benchmarking

- Ensure that your evaluation metrics are appropriate and that you are comparing your method against suitable benchmarks. Reassess the validity of your experimental setup.

Document and Report Findings

- Be transparent about the challenges you faced and the steps you took to address them. Document your findings, even if the improvements are minimal. This transparency contributes to the scientific community's collective knowledge.

Consider Alternative Approaches

- If, despite your efforts, you are unable to achieve significant improvements, consider exploring alternative research approaches. Sometimes, a change in direction can lead to valuable insights.

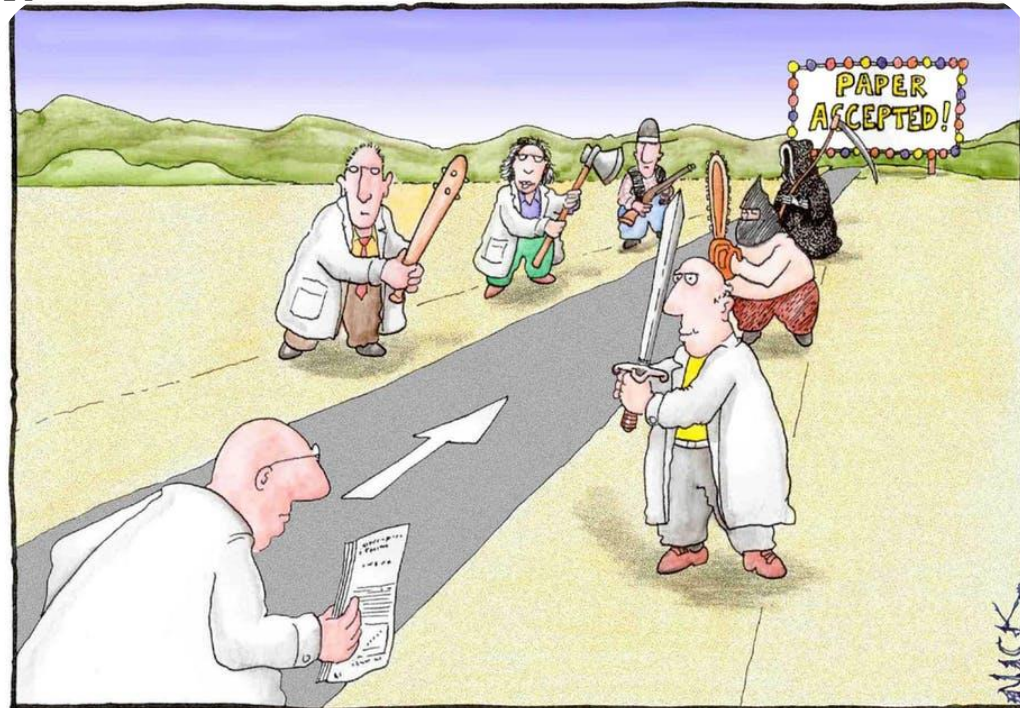
Outline

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- **Study and Implement Existing Methods**
- **How to develop a research plan**
- **Implement Your Proposed Method**
- **Performance Evaluation**
- **Enough Contributions for Submission**



Enough Contributions for Submission?

Getting your conference or journal paper accepted involves a combination of factors, including the quality of your research, writing, and presentation



Most scientists regarded the new streamlined peer-review process as "quite an improvement."

Most scientists regarded the new streamlined peer-review process as "quite an improvement."

Enough Contributions for Submission?

Deciding whether a research paper represents a sufficient contribution for submission to a conference or journal involves assessing the novelty, significance, and completeness of the work

Novelty

Clearly identify and articulate the novelty of your work. Consider whether your research introduces new ideas, concepts, methods, or applications that advance the current state of knowledge in the field.

Literature Review

Ensure that your paper provides a thorough literature review that places your work in the context of existing research. Highlight the gaps in the literature that your work addresses and explain how your contribution fills these gaps.

Research Question and Objectives

Clearly state the research question or problem your paper addresses and outline the specific objectives. Ensure that the objectives align with the overall goal of making a meaningful contribution to the field.

Methodology

Detail the methodology used in your research. Explain the experimental setup, data collection process, and any novel techniques or approaches employed. This section should be comprehensive enough for others to replicate your work.

Enough Contributions for Submission?

Deciding whether a research paper represents a sufficient contribution for submission to a conference or journal involves assessing the novelty, significance, and completeness of the work

Results and Analysis

Present your results in a clear and organized manner. Include visualizations, statistical analyses, or other relevant methods to support your findings. Discuss the implications of your results and how they contribute to the field.

Discussion

Provide a thorough discussion of your results, drawing connections between your findings and the existing literature. Discuss any unexpected outcomes, limitations, and potential future directions for research in your area.

Comparisons and Baselines

If applicable, compare your results with existing methods, baselines, or state-of-the-art approaches. Highlight the strengths and weaknesses of your approach and explain how it advances beyond the current state of the art.

Adherence to Guidelines

Follow the submission guidelines provided by the conference or journal rigorously. Ensure that your paper meets all formatting and length requirements.

Enough Contributions for Submission?

In research, a new and original contribution refers to a novel idea, concept, method, insight, or discovery that advances the current state of knowledge in a particular field

Innovative Approach
or Methodology

- Introducing a *new and innovative research* methodology or approach to solve a problem can be a significant contribution. This could involve developing a *new experimental technique, data analysis method*, or computational algorithm.

Novel Insights or
Observations

- Providing new insights or observations that challenge existing theories or expand our understanding of a phenomenon is a valuable contribution. This might involve *uncovering unexpected patterns, relationships, or trends in data*.

Development of New
Theories or Models

- Proposing a *new theoretical framework* or model that *explains phenomena more accurately* or comprehensively than existing models contributes to the theoretical foundation of a field.

Enough Contributions for Submission?

In research, a new and original contribution refers to a novel idea, concept, method, insight, or discovery that advances the current state of knowledge in a particular field

Creation of New
Tools or
Technologies

•Developing *new tools, technologies, or instruments* that enable researchers to *explore, measure, or analyze phenomena* in ways not previously possible can be a groundbreaking contribution.

Synthesis of
Existing Knowledge

•*Integrating and synthesizing existing knowledge from disparate sources to generate a new conceptual framework* or theory can be a valuable contribution. This involves bringing together diverse ideas in a coherent and meaningful way.

Application in a
Novel Context

•Applying *existing theories, methods, or technologies to a new and unexplored context can be considered a contribution*, especially if it yields insights that were not apparent in the original context.

Enough Contributions for Submission?

In research, a new and original contribution refers to a novel idea, concept, method, insight, or discovery that advances the current state of knowledge in a particular field

Addressing an Unmet
Need or Problem

• *Identifying and addressing a previously unmet need or problem in a field can be a significant contribution.* This might involve solving practical challenges, *improving efficiency*, or addressing gaps in current knowledge.

Verification or
Refutation of
Existing Findings

• Conducting rigorous experiments or studies to *verify* or *refute existing findings* is a valuable contribution. This adds to the *reliability* and *robustness* of knowledge in a given area.

Cross-Disciplinary
Connections

• Making *connections* between *seemingly unrelated fields* and *applying concepts or methods from one discipline to another* can lead to innovative solutions and contribute to the advancement of both fields.

